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Efficient and explainable multivariate time series forecasting: a survey of architectures, taxonomies, and open challenges

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Efficient and Explainable Multivariate Time Series Forecasting: A Survey of Architectures, Taxonomies, and Open Challenges

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Abstract

The rapid expansion of multivariate time series (MTS) data has made deep learning a central tool for forecasting across industrial and scientific domains. As these architectures move into settings such as clinical decision support, industrial monitoring, and financial risk management, researchers need to examine not only predictive accuracy but also computational cost and the kinds of explanations that a model can support. This survey re-examines deep learning models for MTS forecasting through the requirements of efficiency and explainability. Our contributions are threefold: (1) We synthesize the literature through a dual-axis scalability view over sequence length $\boldsymbol{L}$ and variate dimension $\boldsymbol{N}$, together with a three-question explainability taxonomy covering temporal importance, variate importance, and pattern decomposition. (2) We review dominant architectural families, including Transformer variants, frequency-domain methods, linear and channel-independent backbones, state space models, and foundation-style forecasters, to describe how different designs occupy different efficiency–explainability profiles under specific data and deployment assumptions. (3) Building on this synthesis, we identify key open challenges—including the absence of standardized explainability benchmarks for time series, the interpretability gap in state space models, and the need to advance from correlational to causal explanations—and offer reporting considerations to guide more comparable future research.

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Keywords: multivariate time series, deep learning, explainability, efficiency, forecasting

1 Introduction

Deep learning has established itself as an important tool for multivariate time series (MTS) forecasting, supporting decision-making across a broad range of industrial and scientific domains. In intensive care, models use multi-channel physiological signals—including heart rate, blood pressure, and oxygen saturation—to anticipate adverse events Vimbi et al. (2024). Financial applications combine market indices and trading volumes for forecasting and anomaly detection Arsenault et al. (2024); Wang et al. (2022), while industrial systems draw on temperature, pressure, vibration, and energy measurements for maintenance and operational planning Shaikh et al. (2022); Wang et al. (2022). Long Short-Term Memory networks (LSTMs) Lai et al. (2018) and Transformer-based models Vaswani et al. (2017); Zhou et al. (2020, 2022); Wu et al. (2023) have substantially expanded forecasting capabilities in these settings, where predictive performance has traditionally been the principal measure of success.

An accuracy-centered view, however, becomes insufficient when trust and deployability matter alongside precision. Two related challenges illustrate this limitation. *First*, opaque forecasts are difficult to examine in high-risk decisions: when a model predicts elevated clinical risk, a user may need to know whether cardiac, respiratory, or other signals contributed to that result Vimbi et al. (2024); Gandin et al. (2021). Existing time-series XAI surveys also report limited agreement on domain-specific evaluation protocols Rojat et al. (2021); Arsenault et al. (2024). Post-hoc methods can provide useful evidence, but gradient evaluation, perturbation sampling, surrogate fitting, and counterfactual optimization introduce additional computational cost. *Second*, resource-intensive models can be difficult to deploy on embedded or real-time systems Saheed et al. (2024); in particular, the $O(L^2)$ attention cost of a vanilla Transformer limits scalability to long sequences Vaswani et al. (2017). These concerns motivate a joint analysis of efficiency and explainability rather than the assumption that accuracy necessarily requires opacity or that lightweight models are automatically interpretable.

The form of explanation required is also task-dependent. A clinical alarm may call for both the responsible physiological channel and the time window in which deterioration emerged; industrial maintenance may require sensor-level attribution linked to a trend or periodic fault signature; and a latency-sensitive financial system may permit only a compact explanation produced within a strict deadline. A model that exposes seasonal components may therefore be informative for one decision yet insufficient for another that requires cross-variate attribution.

Addressing these challenges requires a broader view of efficiency in time-series modeling. We assess how computational cost scales with both sequence length L and variate count N, since temporal compression does not necessarily reduce the cost of cross-variate interaction. Transformer variants commonly retain $O(L^2)$ or $O(L \log L)$ temporal operations Zhou et al. (2020, 2022); Wu et al. (2023); Eldele et al. (2024),

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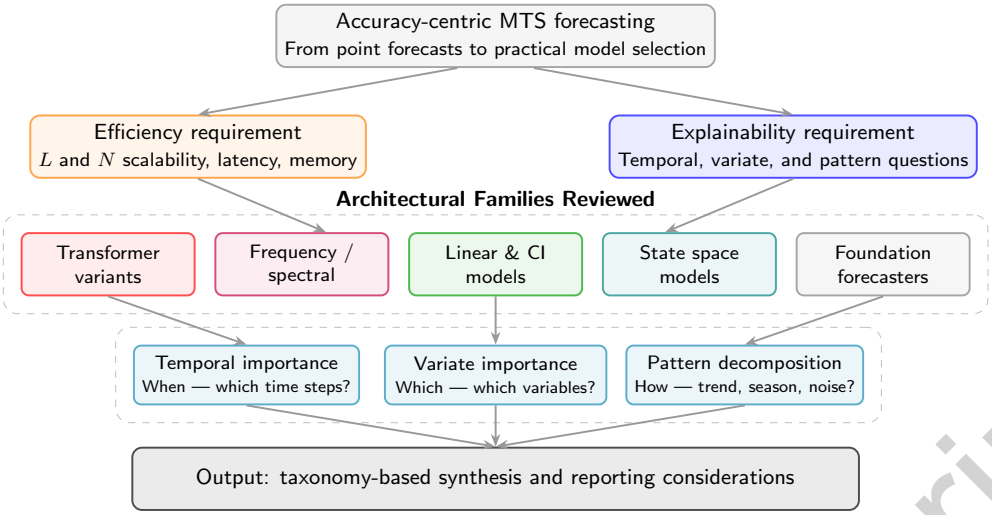


Fig. 1 Overview of the survey's analytical structure. Efficiency is assessed with respect to sequence length L, variate count N, latency, and memory, while explainability is organized around temporal importance, variate importance, and pattern decomposition. These two requirements guide the review of Transformer, frequency-domain, linear and channel-independent, state-space, and foundation-style model families.

whereas linear and frequency-based models can approach $O(L)$ or $O(L \log L)$ temporal scaling Zeng et al. (2023); Xu et al. (2024); Nam et al. (2024). FITS, for example, reports competitive forecasting with approximately 10K parameters through frequency-domain interpolation in its experimental setting Xu et al. (2024). A model that is efficient in L, however, can remain expensive in N when it applies dense cross-variate attention. We therefore consider asymptotic scaling together with empirical indicators—FLOPs, parameter count, memory footprint, and inference latency—whose realized values depend on constant factors, batching, software, and hardware Campos et al. (2023); Han et al. (2024); Luo and Wang (2024).

These measures also refer to different stages of use: training cost affects experimentation and model refresh, inference cost constrains deployment, and explanation cost arises when an explanatory output is requested. Tokenization changes the effective problem size as well, because patching reduces the number of temporal tokens, inverted designs shift attention toward N, and spectral transforms move part of the computation into a structured domain. Accordingly, we consider parameter count together with latency, memory, and explanation overhead.

The second requirement concerns model transparency. We use *interpretability* for transparency supplied by a model's internal structure, including inspectable linear weights, explicit trend–seasonal decomposition, or operations linked to named concepts Xu et al. (2024); Nam et al. (2024); Oreshkin et al. (2019); Challu et al. (2022); Zhou et al. (2022). We use *explainability* for mechanisms that expose or approximate reasons for a prediction from an otherwise opaque model, including SHAP Arsenault et al. (2024), LIME Guilleme et al. (2019), Integrated Gradients, and perturbation tests. Attention occupies an intermediate position: it exposes an architecture-internal

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signal whose value as attribution depends on its agreement with the fitted model's behavior Bastings and Filippova (2020); Vashishth et al. (2019). Post-hoc methods also add costs whose magnitude depends on the explainer, input dimensions, batching, implementation, and hardware.

For MTS forecasting, we organize explanation support around three questions. *Temporal importance* asks which past time steps influence a forecast Li et al. (2023); Eldele et al. (2024); *variate importance* asks which input channels contribute Zhang and Yan (2023); Cirstea et al. (2022); Guo et al. (2023); and *pattern decomposition* asks whether trend, seasonal, spectral, or residual components shape the output Oreshkin et al. (2019); Zhou et al. (2022); Liu et al. (2022). Here, *transparency* is the broader availability of inspectable model and data-flow information; *faithfulness* is agreement between an explanation and the fitted predictor's behavior; and *causality* concerns intervention-level claims under explicit structural assumptions. A representation can therefore be transparent without being faithful, and an attribution can be faithful to a predictor without being causal. Figure 1 summarizes how these questions are used alongside the efficiency dimensions.

These questions define the scope of an explanation, while faithfulness, stability, and usefulness characterize its quality. Each can be answered globally, by summarizing behavior across a dataset, or locally, for one forecast and horizon. A model developer may need sensitivity and failure diagnostics, whereas a domain user may need a stable account at an appropriate temporal resolution. We discuss uncertainty as a complementary requirement, especially for probabilistic forecasts, in Section 3.

Despite the breadth of these individual reviews, an important gap remains: few surveys examine computational efficiency and model transparency through a shared architectural lens. Rojat et al. (2021) established foundational taxonomies for time-series XAI, but its emphasis on CNN and RNN architectures predates the widespread adoption of Transformers for forecasting Zhou et al. (2020); Wu et al. (2021). Arsenault et al. (2024) provided domain-specific insights for financial forecasting while focusing primarily on post-hoc methods. Zhang et al. (2025) reviewed frequency-domain methods from classical Fourier analysis to neural operators and identified open questions in causal structure, uncertainty, and non-Euclidean data, but did not center the relationship between efficiency and explainability. Complementing these surveys, Wang et al. (2024b) introduced TSLib, a unified library covering 30 models and 30 datasets, highlighting reproducibility problems caused by inconsistent splits, preprocessing, and hyperparameter tuning. Other reviews cover Fuzzy Cognitive Maps Orang et al. (2022), bibliometric trends Saputro et al. (2023), anomaly localization Lee et al. (2023), Transformer variants, and frequency-domain forecasting Yi et al. (2024); Hasan et al. (2023); Somvanshi et al. (2024); Yi et al. (2023b,a, 2024); Wang et al. (2025). Together, they provide rich coverage of individual methods while leaving room for a systematic synthesis of how architectural choices shape both computational cost and explanation support.

Accordingly, the primary contribution of this paper is a systematic survey of efficient and explainable MTS forecasting. The dual-axis efficiency view, three-question taxonomy, retained-evidence catalog, illustrative faithfulness analysis, and reporting considerations provide a common structure for organizing and comparing the literature.

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The central thesis of the survey is that the efficiency–explainability relationship is not a single fixed trade-off, but a set of architecture-dependent tensions across temporal, variate, and pattern-level questions. Linear and decomposition-based designs can align low temporal cost with temporal or pattern transparency Xu et al. (2024); Zeng et al. (2023), while channel-independent processing confines that transparency to within-variate structure. More complex architectures offer broader interaction modeling but may add explanation cost. Causality-aware models such as CUTS Cheng et al. (2023), sparse-attention mechanisms Lin et al. (2024); Lu et al. (2024), and hybrid temporal-frequency architectures Chen et al. (2023) offer different combinations of computational cost and explanation support. We therefore compare model families according to the explanatory questions they address, the assumptions behind their signals, and the evidence used to validate them.

The remainder of this survey is organized as follows. Section 2 outlines our survey methodology, including search strategy and study selection criteria. Section 3 formalizes our evaluation framework, defining dual-axis efficiency metrics and multi-dimensional interpretability taxonomies. Section 4 systematically reviews contemporary DL-MTS models, mapping them onto an efficiency-explainability roadmap and analyzing architectural evolution across dominant paradigms. Section 5 examines how causal discovery, adaptive mechanisms, and foundation-scale modeling reshape the trade-offs. Section 6 synthesizes open challenges and future research directions, while Section 7 provides concluding remarks.

2 Survey Methodology

This survey adopts a systematic approach to ensure comprehensive coverage and reproducibility of the literature selection process. The methodology follows established guidelines for conducting systematic reviews in software engineering and artificial intelligence research, adapted for the interdisciplinary nature of multivariate time series (MTS) forecasting with deep learning.

2.1 Search Strategy

The literature search was conducted across four major academic databases: Web of Science (WoS), IEEE Xplore, ACM Digital Library, and arXiv. These databases were selected to capture the full spectrum of relevant publications, from peer-reviewed journal articles and top-tier conference proceedings to rapidly evolving preprint research that characterizes the deep learning community.

The primary search was executed using the following boolean query, applied to titles, abstracts, and keywords:

```
("Multivariate Time Series" OR "MTS Forecasting" OR "Long-term Time Series
Forecasting") AND ("Explainable AI" OR "Interpretability" OR "Attention
Mechanism" OR "State Space Models" OR "Computational Complexity" OR
"Efficient Forecasting")
```

The primary database search was frozen in June 2025. The normalized query above was translated into database-specific syntax as follows: WoS Topic fields (title,

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Table 1 Comprehensive Comparison of Recent Surveys on Deep Learning for Time Series Analysis

Survey	Year	Venue	Scope (Data)	Key Perspective	Efficiency	Explainability
Architecture & Forecasting						
Wen et al. Wen et al. (2022)	2022	IJCAI	LTSF (Elec., Weather, Traffic)	Attention (Sparse/Low-rank); Decomposition	**High** ($O(N \log N)$)	Medium
Liang et al. Liang et al. (2024)	2024	KDD	General TS; Zero-shot	Foundation Models; LLM-based Adaptation	Medium	Medium
Jin et al. Jin et al. (2024)	2024	TPAMI	MTS Forecasting, Anomaly, Imputation	GNN4TS: Spatial + Temporal Graph Modeling	**High** (GNN)	Medium
Patro et al. Patro et al. (2024)	2024	arXiv	Long Sequence Modeling	SSM: from S4 to Mamba; Selective Mechanisms	**High** ($O(N)$)	Low
Patro et al. Patro and Agneeswaran (2024)	2024	arXiv	Long Sequence; Multi-domain	SSM as Transformer Alternative: Mamba	**High** ($O(N)$)	Low
Explainability & Interpretability						
Arsenault et al. Arsenault et al. (2024)	2024	ACM CS	Financial TS Forecasting	XAI Taxonomy: Interpretable vs Explainable	Low	**High**
Vimbi et al. Vimbi et al. (2024)	2024	Cog. Comp.	General ML	Post-hoc XAI: LIME, SHAP, Attention	Low	**High**
Vollert et al. Vollert et al. (2021)	2021	IEEE ETFA	Predictive Maintenance	Interpretable ML: Model-agnostic vs Built-in	Low	**High**
Assaad et al. Assaad et al. (2022)	2022	JAIR	MTS Causal Discovery	7 Method Families: Granger/Constraint/Noise	Low	Medium
Hasan et al. Hasan et al. (2023)	2023	TMLR	I.I.D. + TS Causal Discovery	Causal Discovery Taxonomy	Low	Medium
Specialized Topics						
Yi et al. Yi et al. (2024)	2024	arXiv	MTS Forecasting	Frequency-domain MLPs (DFT/DWT/Wavelet)	**High** ($O(N \log N)$)	Medium
Zhang et al. Zhang et al. (2025)	2025	arXiv	TS Analysis	Classic Transforms to Neural Spectral Operators	**High**	Low
Somvanshi et al. Somvanshi et al. (2024)	2024	ACM CS	General ML	Kolmogorov-Arnold Network (KAN)	Medium	Medium
Iglesias et al. Iglesias et al. (2023)	2023	NCA	TS Classification, Forecasting	Data Augmentation: Transform/Pattern/Gen.	Low	Low
Jin et al. Jin et al. (2024)	2024	TKDE	Spatio-Temporal Urban Data	STGNN: Graph Spatial + Temporal	**High** (GCN+RNN)	Medium

Efficiency: **High** = explicitly optimizes complexity ($O(N)$–$O(N \log N)$); Medium = discusses but not core; Low = not addressed. **Explainability**: **High** = XAI is core topic; Medium = includes interpretability analysis; Low = not addressed. **Venue**: ACM CS = ACM Computing Surveys, NCA = Neural Computing & Applications, Cog. Comp. = Cognitive Computation, TMLR = Trans. Mach. Learn. Res., JAIR = J. Artif. Intell. Res., IEEE ETFA = IEEE Int. Conf. Emerging Technol. Factory Autom.

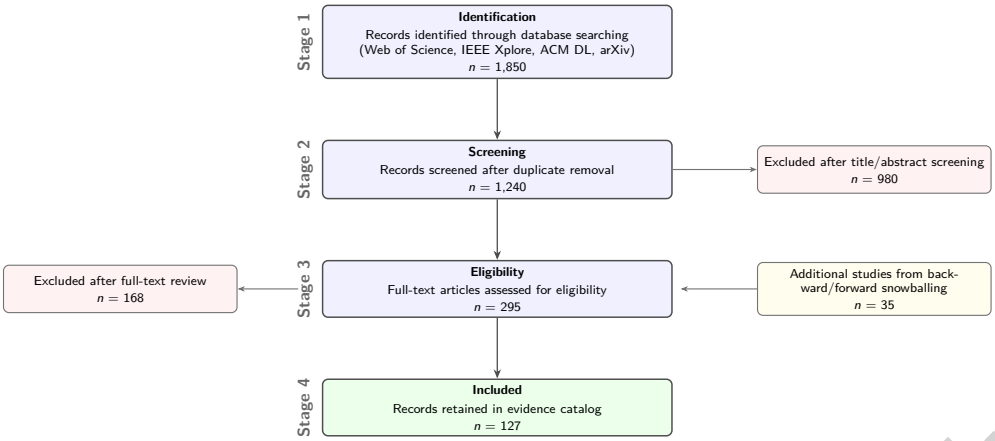


Fig. 2 PRISMA flow diagram of the study selection process. The systematic search across four databases yielded 1,850 initial records. After deduplication and title/abstract screening, 260 candidate records remained; an additional 35 records were added through backward/forward snowballing, yielding 295 full-text assessments and 127 retained evidence records.

abstract, author keywords, and Keywords Plus), IEEE Xplore metadata fields, ACM Digital Library title/abstract/keyword fields, and arXiv title/abstract fields. For WoS, IEEE Xplore, and ACM, we retained English-language journal articles, conference papers, and early-access records within the target period. For arXiv, we retained preprints that satisfied the influence threshold in I3 or were identified through expert snowballing as necessary to explain a major architectural transition. This rule provided timely coverage of 2024–2025 developments while retaining a clear distinction between peer-reviewed work and recent preprints.

The primary search window spanned from January 2020 to June 2025, capturing the period of rapid architectural innovation that followed the introduction of Transformer-based forecasting models. To identify relevant studies beyond the primary query, we used two complementary retrieval strategies: (1) backward snowball citation tracking from influential works in the field Vaswani et al. (2017); Zhou et al. (2020); Zeng et al. (2023), and (2) forward citation analysis using Semantic Scholar's API to identify subsequent studies that built upon these architectures. A narrow post-freeze expert-update pass then added directly relevant architecture and explainability studies needed to contextualize emerging concept-based and causal-pattern explanations. Supplementary Table S1 identifies these records and documents their role in the synthesis.

Supplementary Table S1 assigns every retained item an explicit `record_type` and `inclusion_reason`. Model-facing records cover forecasting architectures; model-facing transfer or contextual records provide directly reusable mechanisms or architectural context; supporting anchors define XAI methods, benchmark protocols, or survey background; and expert-update records identify the narrow post-freeze additions. These roles distinguish the retained evidence catalog from the full reference list, which also contains reporting, regulatory, and general methodological citations.

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2.2 Inclusion and Exclusion Criteria

To ensure the relevance and quality of the final corpus, a set of predefined inclusion and exclusion criteria was applied during the screening process.

Inclusion Criteria

Records were included if they satisfied the following role-aware conditions:

- **I1.** The record either (a) proposes or evaluates a deep learning architecture for time series forecasting, with publication date between 2020 and 2025, thereby capturing the period of rapid architectural evolution from early Transformers to contemporary state space models, or (b) provides a foundational architecture, XAI, causal, benchmark, or survey anchor required to define the framework. Records in group (b) are coded as supporting anchors in Supplementary Table S1.
- **I2.** The study explicitly analyzes at least one of the following: (a) theoretical computational complexity (e.g., $O(L)$, $O(L \log L)$, $O(L^2)$), (b) empirical efficiency metrics (FLOPs, parameter count, inference latency), or (c) interpretability mechanisms—whether intrinsic (e.g., attention weights, linear decomposition, basis functions) or post-hoc (e.g., SHAP, LIME, gradient-based attribution).
- **I3.** Primary model records are published in top-tier conferences (e.g., NeurIPS, ICML, ICLR, AAAI, KDD, IJCAI), top-tier journals (e.g., IEEE TPAMI, IEEE TKDE, Nature Machine Intelligence, Artificial Intelligence Review), or are widely cited arXiv preprints ($\geq$50 citations) that have demonstrably influenced subsequent published work. Supporting anchors must either be peer-reviewed, widely adopted methodological references, or reporting/benchmark standards necessary to interpret the framework.

Exclusion Criteria

Studies were excluded if any of the following conditions applied:

- **E1.** The study focuses strictly on univariate time series (UTS) forecasting without proposing architectures generalizable to the multivariate setting, as the cross-variate modeling dimension is central to this survey's evaluation framework.
- **E2.** The study presents purely black-box optimization techniques (e.g., neural architecture search, hyperparameter tuning) without providing computational cost analysis or interpretability assessment, as such work falls outside the efficiency-explainability scope of this review.
- **E3.** The study is a non-peer-reviewed preprint with fewer than 50 citations and no demonstrable influence on subsequent published research.

Adjacent time-series tasks were screened according to their relevance to forecasting. Studies on anomaly detection, imputation, classification, spatiotemporal forecasting, probabilistic forecasting, or graph-based temporal modeling were included when they contributed a forecasting architecture, a transferable MTS representation mechanism,

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or a directly relevant efficiency/explainability analysis. Papers confined to task-specific detection, classification, or domain engineering were excluded even if they used time-series inputs. This rule kept the corpus focused on mechanisms that inform forecasting model design.

2.3 Study Selection Process

The study selection process followed PRISMA guidance Page et al. (2021). The initial search yielded approximately 1,850 records. We deduplicated by DOI, arXiv identifier, normalized exact title, and manual review of near-duplicate titles with overlapping authors. When a preprint and a later peer-reviewed version described the same work, the peer-reviewed version was the primary record; the preprint was consulted only for implementation details absent from the publication. This produced 1,240 unique records. Title and abstract screening retained 260 candidates, and snowballing added 35 records for a total of 295 full-text assessments. Excluding 168 records yielded the 127-record evidence catalog shown in Figure 2.

The task coding comprises 91 model-facing, transfer, contextual, or expert-update records (68 forecasting/MTS modeling, 5 causal forecasting/explanation, 4 anomaly-detection transfer, 2 classification transfer, and 12 contextual architecture/motivation records) and 36 supporting anchors (22 XAI/interpretability methods, 11 survey/background records, and 3 benchmark/evaluation protocol records). Supplementary Table S1 reports the citation key, title, year, venue or source, record type, inclusion reason, task type, multi-label architecture family, complexity class, explanation mechanism, inclusion source, catalog-scope role, and verification status for every retained record. It is uploaded as a separate supplementary file so that the coding and corpus boundary can be inspected. References used only for reporting standards, regulation, general deep-learning background, or other context are not counted in the 127-record catalog.

Two independent reviewers (the first and second authors) screened and classified the records. Initial inclusion agreement was 92% (Cohen's $\kappa = 0.84$); a third senior reviewer helped resolve all disagreements. Architecture families use multi-label coding when a model spans, for example, decomposition and Transformer mechanisms, frequency and channel-independent processing, or graph and attention components.

3 Evaluation Framework

Having established the systematic literature selection process in Section 2, we now introduce the analytical lens used to synthesize the selected models. The rapid diversification of MTS forecasting architectures—from Transformer variants and frequency-domain filters to linear and state-space models—calls for a common basis on which their computational and explanatory properties can be discussed. We use a dual-axis view of efficiency, distinguishing sequence length L from variate count N, together with three explanation questions: when, which, and how a forecast is explained. The resulting framework organizes the model properties and empirical findings reported in the cited studies. Figure 3 connects this evaluation lens to the model-family synthesis and emerging-paradigm discussion.

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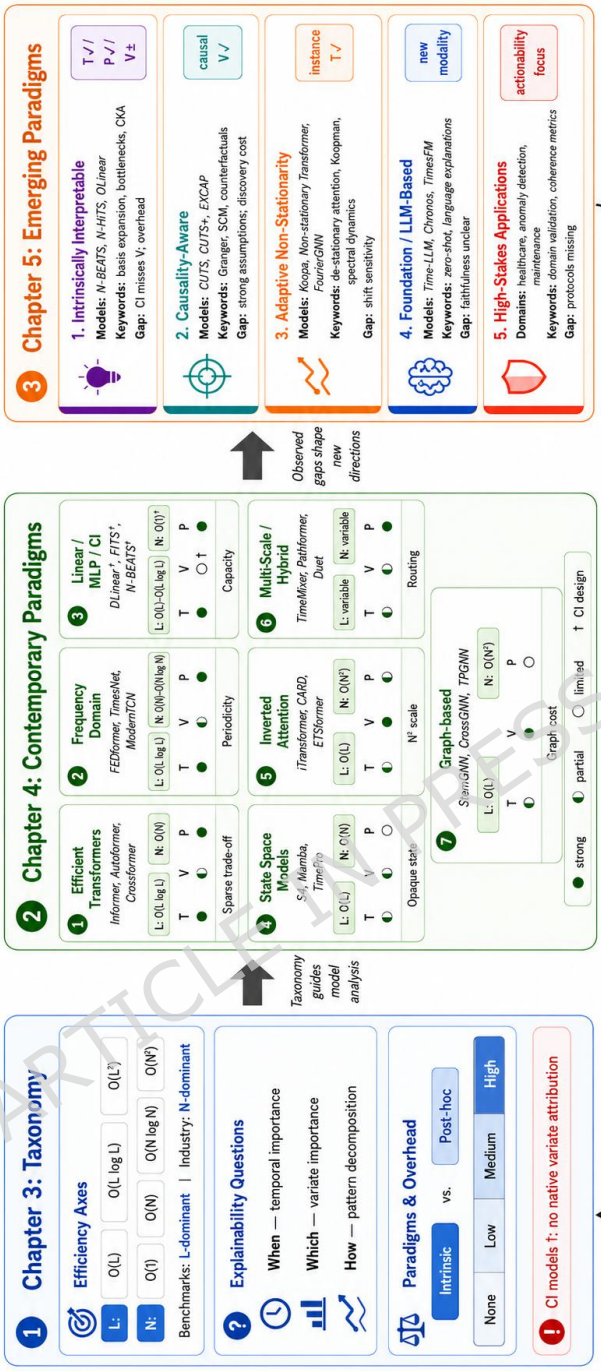


Fig. 3 Survey roadmap linking the evaluation lens in this section to the model-family synthesis in Section 4 and the emerging directions in Section 5.

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3.1 Efficiency Metrics for Time Series Forecasting

The practical utility of a forecasting model is often constrained by its computational footprint. As sequence lengths and data dimensionality grow, efficiency becomes a decisive factor. This subsection details the metrics used to quantify model efficiency, spanning from theoretical complexity analysis to empirical performance benchmarks, which together offer a holistic view of a model's operational cost.

3.1.1 Theoretical Computational Complexity

Theoretical complexity provides a hardware-agnostic measure of how a model's computational requirements scale with the size of its input. For multivariate time series forecasting, two primary scaling dimensions must be distinguished:

- **Sequence length (L):** The number of historical time steps in the input window. This dimension dominates attention complexity in temporal self-attention ($O(L^2)$).
- **Variate count (N):** The number of input channels or variables. Models like iTransformer Liu et al. (2024) that apply attention across variates scale as $O(N^2)$ in the variate dimension while remaining linear in L.

While the majority of prior surveys focus almost exclusively on the temporal dimension L, the variate dimension N introduces an equally important—and in high-dimensional settings, potentially dominant—scaling axis that warrants systematic analysis. Contemporary MTS models exhibit four distinct complexity regimes along the N dimension:

1. **Channel-independent (CI), $O(1)$ per variate:** Models such as DLinear Zeng et al. (2023), FITS Xu et al. (2024), and PatchTST Nie et al. (2023) process each variate through an identical but independent pathway, yielding a total complexity of $O(L{\cdot}N)$ (or $O(L \log L{\cdot}N)$ for frequency-domain CI models). This design favors temporal efficiency and confines its native explanatory scope to within-variate structure.
2. **Linear cross-variate mixing, $O(N)$:** Models including Informer Zhou et al. (2020), Autoformer Wu et al. (2021), FEDformer Zhou et al. (2022), TimesNet Wu et al. (2023), S-Mamba Wang et al. (2024), and TSMixer Ekambaram et al. (2023) incorporate cross-variate interaction through linear embedding layers, feature mixing MLPs, or linear-scan state transitions. These mechanisms enable implicit variate fusion at linear cost in N, though the resulting cross-variate representations are typically entangled and difficult to interpret.
3. **Quasi-linear cross-variate, $O(N \log N)$:** Frequency-domain architectures such as the Fourier Neural Filter (FNF) Xu et al. (2025) perform cross-variate mixing via FFT operations along the channel dimension, achieving $O(N \log N)$ scaling with structured spectral interpretability.
4. **Quadratic cross-variate, $O(N^2)$:** Models that apply full attention across variates—notably iTransformer Liu et al. (2024) and Crossformer Zhang and

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Yan (2023)—scale quadratically in N. This cost is often acceptable for low-dimensional benchmarks (e.g., ETTh with $N = 7$), but becomes prohibitive for high-dimensional industrial datasets such as the PEMS traffic series ($N = 170$–883) or hospital monitoring systems with hundreds of physiological channels.

The practical significance of N-dimension scaling is dataset-dependent: on common academic benchmarks where $L = 512$ and $N = 7$–21, the L dimension may dominate and differences in N-scaling can be difficult to observe. In deployments involving sensor networks ($N > 100$) or financial portfolios ($N > 500$), however, the variate dimension can become a binding constraint. For PEMS08 ($N = 170$), a full cross-variate attention layer contains 28,900 pairwise interaction terms, whereas a linear-in-N mechanism scales with 170 variate positions. The measured latency and memory ratio also depends on kernel choice, hidden dimension, batching, and hardware. Table 8 therefore reports both L- and N-complexity alongside protocol-specific empirical evidence.

Throughout this survey, we report complexity with respect to L unless otherwise noted; variate-dimension scaling is indicated explicitly in text and tables. For models with mixed scaling, we report the dominant term for typical MTS configurations where $L \gg N$ (e.g., $L = 512$, $N = 7$–21), while flagging cases where N-complexity becomes dominant.

Canonical Transformer architectures exhibit $O(L^2)$ complexity because self-attention compares all token pairs Vaswani et al. (2017). Frequency-domain models can reduce temporal scaling to $O(L \log L)$ Xu et al. (2024): FEDformer retains this FFT complexity while selecting a sparse set of M Fourier modes after transformation to reduce the practical cost of frequency mixing Zhou et al. (2022). State-space models pursue $O(L)$ scans without full attention. These classes are useful for describing scalability, but asymptotic complexity alone does not determine wall-clock performance: constant factors, memory access, batching, and hardware kernels can reverse an expected ordering. Table 2 therefore separates theoretical complexity from empirical indicators.

Table 3 summarizes recurring patterns in the reviewed studies. Because look-back windows, horizons, normalization, hardware, and hyperparameter budgets differ across papers, the table supports a qualitative family-level comparison. The available evidence shows that lightweight models can be competitive in suitable settings and illustrates the value of shared protocols, confidence intervals, and comparable explanation metrics.

We distinguish three levels of comparison. *Within-study comparisons* use results obtained with a common dataset split, preprocessing pipeline, horizon, implementation environment, and tuning budget. *Cross-study summaries* preserve the values, conditions, and complexity claims reported by their sources. *Family-level synthesis* compares architectural mechanisms, asymptotic scaling, and explanation coverage qualitatively. This structure keeps empirical results connected to their experimental settings and explanatory mechanisms connected to their architectural roles.

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Table 2 Efficiency Metrics for MTS Forecasting Models

Metric Category	Metric	Unit	Scope
Theoretical	Time Complexity	$O(L^n)$	Scalability
	Space Complexity	$O(L^n)$	Memory
	Algorithmic Order	—	Asymptotic
Empirical	FLOPs	GFLOPs	Compute cost
	Inference Latency	ms	Real-time
	Training Time	hours	Development
	Model Size	MB	Deployment

Note: L denotes input sequence length; N denotes variate count. Theoretical complexity provides asymptotic scalability guarantees, while empirical metrics quantify actual deployment costs. Platform-dependent metrics (FLOPs, latency, memory) vary significantly across CPU, GPU, and edge devices, necessitating hardware-specific benchmarking.

Table 3 Qualitative comparison of recurring performance, efficiency, and explanation patterns across model paradigms in the reviewed studies.

Family	Reported performance pattern	Efficiency profile	Comparison caveat
Transformer variants	Strong results are reported across long-horizon and cross-variate tasks, with substantial variation by architecture and dataset.	$O(L \log L)$–$O(L^2)$; flexible long-range and cross-variate modeling	Attention design, patching, and N-scaling differ; results are protocol-dependent.
Linear and decomposition models	Competitive results are reported on several long-horizon benchmarks, especially where trend and periodic structure dominate.	$O(L)$ or $O(L \log L)$; compact temporal or pattern structure	Channel independence can omit cross-variate effects; nonlinear capacity varies.
State-space models	Competitive results are reported for long sequences and some high-dimensional datasets, but the forecasting evidence base is newer.	Usually $O(L)$; compact state propagation	Kernel implementation, channel mixing, and state interpretation vary across models.

Cross-study entries retain the datasets, lookback windows, horizons, preprocessing, hardware, and hyperparameter budgets reported by their sources.

3.1.2 Empirical Performance Indicators

While theoretical complexity offers insight into algorithmic scalability, empirical indicators provide concrete measurements of a model's performance on specific hardware. Three key metrics are central to this evaluation: Floating Point Operations (FLOPs), latency, and memory usage.

FLOPs quantify arithmetic workload, latency measures wall-clock inference time, and memory covers parameters and intermediate activations. For example, Xu et al.

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(2024) reports 4.5K–16K parameters and 0.6 ms GPU inference under its experimental setting. Dual-domain normalization has also been reported to improve stability on non-stationary data with limited added cost Dai et al. (2024b). These values remain hardware- and implementation-dependent and should be compared only when measurement conditions are compatible.

A reproducible empirical efficiency report should specify the input length, variate count, forecast horizon, batch size, numerical precision, warm-up procedure, number of timed repetitions, and whether data transfer and preprocessing are included. Latency should be accompanied by the processor or accelerator, software stack, and implementation source; peak memory should distinguish parameter storage from activations and temporary workspaces. FLOPs and parameter counts remain useful for diagnosis, but they do not capture sparse-kernel utilization, memory bandwidth, parallelism, or compilation effects. For this reason, the framework treats theoretical and empirical indicators as complementary evidence rather than interchangeable measurements.

3.2 A Taxonomy for Interpretability in Time Series Models

Interpretability, or the ability to explain a model's decision-making process in human-understandable terms, is a cornerstone of trustworthy AI. For MTS forecasting, this involves understanding how a model uses historical data to generate future predictions. We organize the selected literature around three intuitive questions: temporal importance (*When?*), variate importance (*Which?*), and pattern decomposition (*How?*).

Interpretability choices also influence efficiency budgets. Architectures that pursue intrinsic interpretability, such as decomposition stacks or sparse prototype libraries, may impose structural constraints that alter parameter sharing and per-token computation. Conversely, post-hoc analyses add perturbation, gradient, or surrogate-model passes whose latency and memory cost depends on the explainer configuration and how often explanations are requested. This interpretability–efficiency tension motivates reporting explanation overhead separately and considering whether intrinsic or on-demand post-hoc analysis is more appropriate for a given deployment.

3.2.1 Intrinsic vs. Post-hoc Approaches

The pursuit of interpretability in deep learning follows two primary paradigms: intrinsic and post-hoc methods. This distinction, systematically framed by Rojat et al. (2021), separates models that are self-explanatory by design from those that require external tools to analyze a pre-trained "black-box" model.

Intrinsic (or ante-hoc) interpretability is achieved through architectural choices that embed explanatory mechanisms directly into the model. The N-BEATS model, for instance, employs a double residual stack structure with basis function expansion, allowing it to explicitly decompose a time series into trend and seasonality components in a manner analogous to classical statistical methods Oreshkin et al. (2019). Attention mechanisms were also initially considered a form of intrinsic interpretability, though their reliability for this purpose is a subject of ongoing debate. In contrast, post-hoc techniques are model-agnostic tools applied after training. These include methods like

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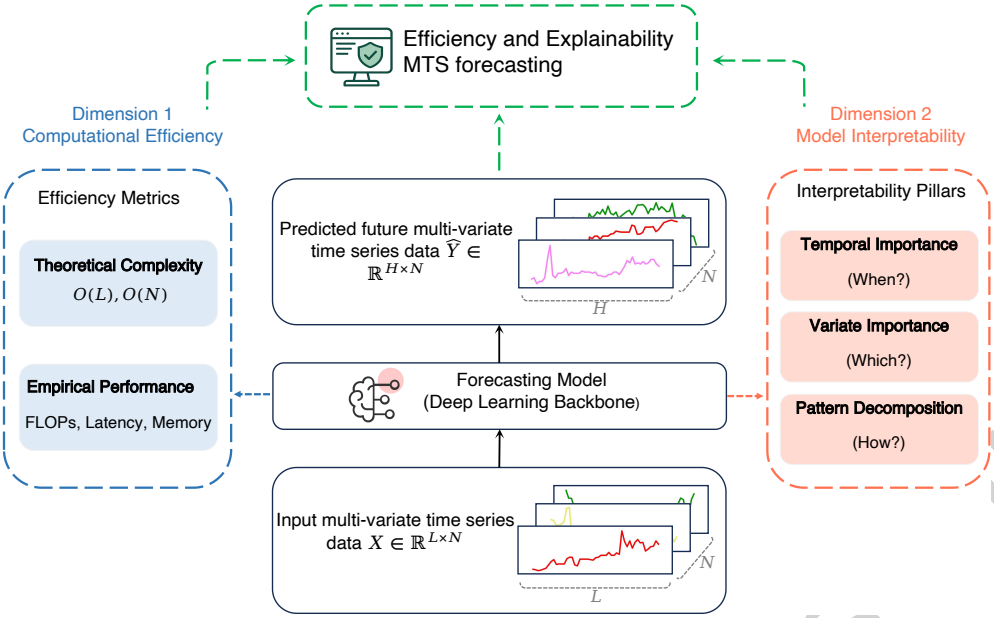


Fig. 4 Evaluation lens used in this survey, combining efficiency measures with three explanation questions: Temporal Importance (*When?*), Variate Importance (*Which?*), and Pattern Decomposition (*How?*).

saliency maps generated via gradients (e.g., Gradient*Input) or perturbation-based approaches like occlusion analysis, which systematically mask parts of the input to measure their impact on the output. With these paradigms clarified, the next subsections examine temporal, variate, and pattern-level attributions to ground the abstract notions of "when," "which," and "how" in concrete analytical tools. Table 4 provides a comprehensive taxonomy of interpretability methods, organized by paradigm (intrinsic vs. post-hoc) and attribution type (temporal, variate, pattern-level), along with their computational costs and the specific explanatory questions they address.

Post-hoc methods bring their own assumptions. In MTS settings, SHAP is sensitive to the background distribution and the treatment of correlated variates; LIME depends on the sampled neighborhood and local surrogate; Integrated Gradients depends on its baseline and integration path; and counterfactual explanations must remain temporally plausible and operationally feasible. Behavioral perturbation, stability, and sensitivity checks help establish how faithfully each method reflects a fitted forecaster.

It is equally important to distinguish an *explanation object* from the claim made about it. An attention matrix, decomposition component, spectral coefficient, graph edge, or concept activation is an inspectable object. Interpreting that object as evidence of influential inputs requires a link to prediction behavior, while a causal interpretation additionally requires explicit structural assumptions and intervention-compatible evidence. The target and granularity should also be stated: an explanation

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for one channel and horizon differs from an attribution averaged across channels, horizons, or a complete dataset. Reporting the target, reference distribution, aggregation rule, and stability across relevant samples makes these distinctions reproducible.

3.2.2 Temporal Importance Attribution

A fundamental question in time series analysis is "When?": which historical time steps were most influential in determining a future prediction? Temporal importance attribution methods aim to answer this by generating a saliency map over the input time axis. This is often achieved by leveraging the attention weights from Transformer-based models, which are presumed to reflect the model's focus on specific past events. However, other techniques exist, such as using the magnitude of gradients with respect to the input or measuring the prediction shift caused by perturbing individual time steps. More advanced models like FEDformer perform attribution in the frequency domain, visualizing sparse attention patterns over Fourier modes to identify critical periodicities Zhou et al. (2022). These visualizations help users understand whether a model is focusing on relevant historical patterns or relying on spurious correlations.

3.2.3 Attention Faithfulness: A Quantitative Perspective

Beyond qualitative assessment, the faithfulness of attention mechanisms—the degree to which attention weights quantitatively reflect the true contribution of input tokens to the model's output—has been a subject of intense debate. This discussion was notably framed by the critical work of Jain and Wallace (2019), who argued that raw attention weights are often poor indicators of feature importance, and the subsequent rebuttal by Wiegreffe and Pinter (2019), which contended that with proper experimental design, attention can provide meaningful explanations. This foundational debate has spurred the development of more rigorous evaluation protocols to measure faithfulness directly.

Several quantitative methods have been proposed to validate attention's explanatory power. These include *attention rollout* Abnar and Zuidema (2020), which aggregates attention weights across layers in Transformer-based models to capture the full flow of information from input to output. Another common approach involves *perturbation tests*, where the correlation between input gradients and attention scores is analyzed to determine whether the most attended-to tokens are also those with the highest gradient magnitude. Furthermore, *leave-one-out (LOO) erasure metrics* systematically remove input tokens and measure the corresponding drop in output probability or accuracy, comparing this change against the initial attention distribution. A faithful attention mechanism should exhibit high correlation between attention weights and the measured importance from erasure tests.

Applying these faithfulness evaluation methods to time series forecasting, however, presents unique challenges. The inherent temporal dependencies and the autoregressive nature of forecasting models mean that the importance of a given timestep can be distributed and propagated through subsequent states, complicating the direct attribution of output contributions to specific input weights. Moreover, time series often exhibit strong autocorrelation, making it difficult to isolate the contribution of individual time steps without violating the distributional assumptions of the data.

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Table 4 Representative architecture-internal and post-hoc explanation methods for MTS forecasting.

Source	Attribution type	Method examples	Extra cost	Question
Architecture-internal	Temporal	Attention weights	$\sim O(1)$	*When?* Relevant historical positions
		Learned decomposition	$\sim O(1)$	*How?* Active trend or seasonal components
	Variate	Cross-attention scores	$\sim O(1)$	*Which?* Candidate influential channels
		Channel-wise gating	$\sim O(1)$	*Which?* Active cross-channel pathways
	Pattern-level	Basis expansion (N-BEATS)	$\sim O(1)$	*How?* Trend or seasonal contribution
		Frequency decomposition	$\sim O(1)$	*How?* Dominant frequency modes
Post-hoc	Temporal	Gradient-based (Grad*Input)	O(backprop)	*When?* Gradient-sensitive positions
		Occlusion/perturbation	$O(L \times \text{inf.})$	*When?* Effect of masking time t
	Variate	SHAP (Shapley values)	$O(2^N)$ (Exact)	*Which?* Variable contribution estimates
		Permutation importance	$O(N \times \text{inf.})$	*Which?* Effect of shuffling variate i
	Pattern-level	LIME (local surrogates)	$O(K \times \text{inf.})$	*How?* Local surrogate behavior
		Counterfactual explanations	$O(I \times \text{inf.})$	*How?* Input change that alters prediction

Note: inf. = inference. Architecture-internal objects impose little extraction overhead; perturbation, stability, or sensitivity tests connect attention and gating scores to forecast behavior.

These challenges underscore the need for time-series-specific faithfulness metrics that account for temporal structure, an area that remains underexplored in the current literature.

Channel-Independent Architectures and Variate Attribution.
Channel-independent (CI) architectures such as DLinear, FITS, and PatchTST process each variate in isolation. Their linear weights or spectral coefficients can expose temporal importance and pattern decomposition within a channel, while cross-variate attribution requires a channel-dependent mechanism or a separate attribution method. We denote this temporal/pattern scope as Level $3^{\dagger}$ in the comparisons below.

3.2.4 Variate Importance Attribution

In the multivariate context, the "Which?" question becomes crucial: which of the input variables (or variates) contributed most significantly to the forecast for a particular target variable? Quantifying variate importance is essential for applications ranging from root cause analysis in industrial systems to portfolio management in finance. Methods for this type of attribution often parallel those for temporal importance. Techniques include analyzing attention weights across variables in multi-head attention layers, aggregating gradient information per variate, or conducting permutation tests where the values of a single variable over time are shuffled to measure the resulting drop in prediction accuracy. An effective model should not only make accurate predictions but also provide a clear and plausible ranking of the input variables that drove its forecast.

Attention-based scores are appealing because they materialize from existing multi-head attention modules and highlight cross-channel interactions with negligible overhead, yet they are sensitive to scaling heuristics (e.g., temperature or head dropout) and can misrepresent causal influence when attention is decoupled from contribution. Gradient-based attributions capture local nonlinear sensitivities and integrate naturally with differentiable training pipelines, but they require smooth preprocessing layers, can saturate when activations are flat, and may fluctuate under input noise. Permutation-based tests are model-agnostic and yield intuitive effect sizes expressed in downstream error metrics, though they demand repeated forward passes and struggle with strongly correlated variates because shuffling breaks realistic joint distributions. Selecting among these approaches therefore depends on whether the priority is low-overhead monitoring, faithful local reasoning, or regulator-grade auditing.

3.2.5 A Quantitative Scoring Protocol for Interpretability

To compare these different forms of explanation more consistently, we use two dimensions: the **Interpretability Level** and the **Explanation Overhead**. Together they describe the kind of explanatory evidence available and the additional computation needed to produce it.

The **Interpretability Level** is rated on a four-point scale (0–3), classifying the nature and faithfulness of the explanation mechanism:

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- **Level 0 (Black Box):** The model offers no specific mechanism to explain its predictions. Standard MLPs or black-box ensembles without post-hoc analysis fall into this category.
- **Level 1 (Qualitative Post-hoc):** The model relies on post-hoc methods, such as gradient or perturbation-based attribution, which are primarily qualitative and may lack theoretical guarantees of faithfulness.
- **Level 2 (Validated Post-hoc):** The model uses a post-hoc method whose output is quantitatively checked against the fitted model's behavior through perturbation, stability, sensitivity, or related faithfulness tests. Level 2 therefore reflects method-specific empirical validation of SHAP, LIME, Integrated Gradients, counterfactual, or related outputs.
- **Level 3 (Intrinsic):** The model's architecture is inherently interpretable, ideally across all three pillars of temporal importance, variate importance, and pattern decomposition. Its structural components—such as basis functions, explicit decomposition modules, or concept bottlenecks—directly provide a transparent mapping from inputs to outputs.

For **channel-independent linear backbones** (e.g., DLinear, FITS, PatchTST), Level $3^{\dagger}$ denotes intrinsic temporal and pattern objects within each variate. Cross-variate attribution falls outside this architecture-internal scope. Attention-based models move from Level 1 visualization to Level 3 intrinsic explanation when their attention weights show strong agreement with behavioral evidence through metrics such as attention rollout Abnar and Zuidema (2020), gradient–attention correlation, or leave-one-out erasure tests Jain and Wallace (2019); Bastings and Filippova (2020).

Evaluative Guidelines and Threshold Calibration.

To illustrate how the framework distinguishes qualitative attention visualization from behaviorally supported explanation, we use a gradient-attention correlation threshold of > 0.7 and a leave-one-out (LOO) agreement threshold of $> 80\%$ as operational reference points for the ETTh1 case study. Calibration across datasets, horizons, and architectures remains an important next step.

The two values form a deliberately demanding joint rule. A correlation above 0.7 requires strong concordance between attention and a gradient-based importance signal, while agreement above 80% requires the attention ranking to be supported by erasure behavior in a large majority of evaluated cases. Correlation and erasure provide complementary checks, as motivated by prior attention-faithfulness studies Serrano and Smith (2019); Wiegreffe and Pinter (2019). The numerical thresholds operationalize this case study and invite future calibration on broader MTS settings.

Domain-Specific Adaptation: The Autocorrelation Caveat.

Strong temporal autocorrelation complicates single-timestep LOO tests because adjacent observations may carry redundant information. If measurements at $t-1$, t, and $t+1$ encode nearly the same local pattern, neighboring values can substitute for an erased value at t, causing single-point erasure to underestimate its contribution. Grouped-window erasure, conditional perturbations, and distribution-preserving

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masks provide useful complementary checks in such settings. The case study therefore combines LOO with gradient–attention correlation and interprets both in light of temporal redundancy.

Illustrative Case Study: Operationalizing the Framework.

We apply the guidelines to two representative attention architectures on an ETTh1 long-horizon forecasting setting ($H = 96$). The case study shows how the thresholds separate qualitative attention visualization from empirically supported explanation. We report mean values with 95% confidence-interval half widths over five random seeds:

- **iTransformer Liu et al. (2024):** By inverting the attention mechanism to operate across the variate dimension (N), iTransformer exposes cross-channel attention maps relevant to the Variate Importance pillar. In the ETTh1 setting, its gradient–attention correlation ($r = 0.466 \pm 0.027$) and LOO consistency ($70.12\% \pm 2.35$ percentage points) fall below the joint Level 3 rule. The result supports a Level 1 classification for these attention maps in this case.
- **Informer Zhou et al. (2020):** The ProbSparse self-attention mechanism selectively attends to a subset of "active" queries. In the same setting, its sparse temporal attention shows weak agreement with gradient and erasure-based signals ($r = -0.128 \pm 0.078$, LOO consistency $20.96\% \pm 0.74$ percentage points), corresponding to Level 1 under the case-study rule.

Table 5 ETTh1 case study applying the attention-faithfulness guidelines to long-horizon forecasting with horizon $H = 96$.

Model	Attention type	Grad.-attn. r	LOO agreement	Outcome
Informer Zhou et al. (2020)	Temporal sparse attention	-0.128 ± 0.078	$20.96\% \pm 0.74$ pp	Below L3
iTransformer Liu et al. (2024)	Variate attention	0.466 ± 0.027	$70.12\% \pm 2.35$ pp	Below L3

Values report mean ± 95% confidence-interval half width over five random seeds. pp denotes percentage points. Outcomes describe the evaluated ETTh1 setting; calibration across datasets, horizons, and checkpoints is left for broader studies.

This tiered assessment separates forecasting efficiency, the availability of attention maps, and demonstrated explanatory fidelity. The five-seed example covers one dataset family, one horizon, and two attention-based models; broader calibration can extend the analysis across datasets, frequencies, checkpoints, and model families.

Complementing this, the **Explanation Overhead** quantifies the additional computational cost required to generate an explanation, measured relative to a single forward pass:

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- **None:** The explanatory object is produced within the forecasting pass, without a separate explainer pass; faithfulness is recorded separately by the Interpretability Level.
- **Low:** The explainer requires one or a few gradient or summary operations beyond forecasting.
- **Medium:** The explainer uses repeated integrations, a bounded number of perturbations, or a model-specific approximation such as DeepSHAP Arsenault et al. (2024).
- **High:** The explainer requires many coalition evaluations, repeated feature or window perturbations, or iterative counterfactual optimization.

Under this rubric, an interpretable N-BEATS configuration can expose architecture-internal pattern components, whereas a post-hoc explanation applied to a black-box forecaster reaches Level 2 only when its output is empirically validated against model behavior. The overhead axis records the additional deployment cost separately from this faithfulness assessment.

Table 6 summarizes the dominant cost drivers and MTS-specific limitations of representative post-hoc methods. Actual latency and memory depend on the implementation, explainer configuration, input dimensions, batching, software, and hardware. Matched measurements with the forecasting model provide the most informative account of this overhead.

Table 6 Qualitative Computational and Faithfulness Considerations for Representative Post-hoc Methods

Method	Main Extra Work	Principal MTS Limitation	Cost Profile
Gradient*Input Simonyan et al. (2013)	Backward gradient evaluation	Local sensitivity can saturate or vary with scaling and noise	Low
Integrated Gradients Sundararajan et al. (2017)	Repeated evaluations along an integration path	Attribution depends on the baseline and path	Medium
Occlusion Analysis Zeiler and Fergus (2014)	Repeated masked forward passes	Masks can be off-distribution; correlated neighbors can substitute for an erased point	Medium–High
SHAP / KernelSHAP Lundberg and Lee (2017a)	Coalition evaluation or surrogate fitting	Results depend on background and conditional assumptions for correlated variates	High
Counterfactuals Wachter et al. (2017)	Iterative constrained optimization	Candidate sequences may be temporally implausible or operationally infeasible	High

Cost profiles are qualitative. Actual latency and memory depend on the implementation, explainer configuration, input dimensions, software, batching, and hardware, and should be measured under a matched forecasting protocol.

Table 7 illustrates how representative models map to the four proposed levels. Attention visualization alone is classified separately from validated attribution, consistent with the concerns raised by Jain and Wallace (2019) and Bastings and Filippova (2020).

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Table 7 Application of Interpretability Scoring Protocol to Representative Models

Model	Level	Overhead	Complexity	Justification
Intrinsically Interpretable (Level 3)				
N-BEATS Oreshkin et al. (2019)	3†	None	$O(L)$	Explicit basis functions; direct coefficient inspection.
N-HiTS Challu et al. (2022)	3†	None	$O(L)$	Hierarchical decomposition with multi-rate sampling.
DLinear Zeng et al. (2023)	3†	None	$O(L)$	Weights give per-channel temporal saliency only.
CPformer Ma et al. (2025)	3	None	$O(L^2)$	5 domain-agnostic concepts; physics-informed constraints.
Attention Evaluated for Faithfulness (Level 1 → 3)				
Informer Zhou et al. (2020)	1*	None	$O(L \log L)$	ETTh1 attention scores fall below the joint case-study thresholds.
iTransformer Liu et al. (2024)	1*	None	$O(N^2)$	ETTh1 variate-attention demo shows partial agreement but remains below Level-3 thresholds.
Post-hoc Explainability (Level 1 → 2)				
TimesNet Wu et al. (2023)	1 → 2	High	Model-dependent	Empirical faithfulness tests determine whether a selected post-hoc method reaches Level 2.

* The ETTh1 case-study measurements place these attention maps at Level 1 under the stated joint rule.
† Channel-independent backbones are marked Level 3 only for temporal/pattern pillars.

3.2.6 Pattern-level Attribution and Decomposition: The "How?" Pillar

Beyond attributing importance to individual time steps or variables, a higher level of interpretability involves understanding how a model captures and utilizes underlying data patterns, such as trend, seasonality, and cycles. This aligns with the classical time series decomposition paradigm. The most prominent example of a deep learning model designed for this purpose is N-BEATS, which explicitly learns to represent and extrapolate trend and seasonal components using dedicated network stacks Oreshkin et al. (2019). Its interpretable configuration was shown to achieve performance on par with its generic counterpart, demonstrating that interpretability need not come at the cost of accuracy. Similarly, recent frequency-domain models provide a different form of pattern-level insight. Zhou et al. (2022) employs a Mixture-of-Experts design for decomposition in the frequency space, while Xu et al. (2024) offers interpretability through the physical meaning of amplitude and phase in its frequency interpolation, as well as visualizations of its low-pass filtering effects. These approaches provide a more holistic, semantic understanding of the model's behavior.

Point and Probabilistic Forecasting.

The evaluation framework above focuses on deterministic point forecasts, which constitute the dominant paradigm in the models surveyed. In high-stakes domains such as clinical decision support and energy grid management, *uncertainty quantification*—through calibrated prediction intervals, conformal prediction sets, or full predictive distributions—is equally important. Explaining *why* a model is uncertain (e.g., whether prediction variance is driven by aleatoric noise in specific variates or epistemic uncertainty from distributional shift) adds a further explanatory dimension. Section 6 discusses probabilistic interpretability as a key direction for extending the framework.

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4 Contemporary Deep Learning Models for MTS Forecasting

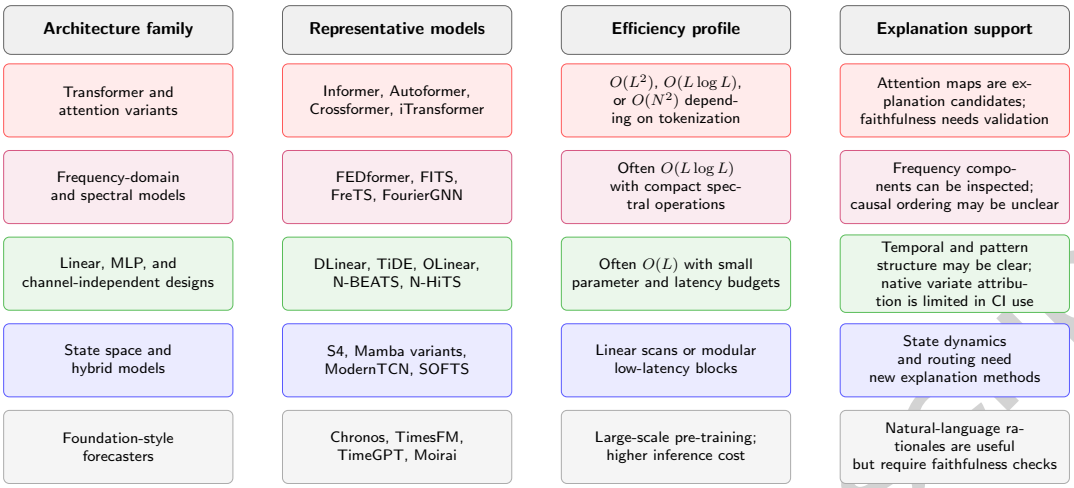


Fig. 5 Five umbrella families of contemporary MTS forecasting architectures. The placement is a qualitative index of representative models, efficiency profiles, and explanation support; finer-grained variants are discussed in the text and Table 9.

Figure 5 provides a compact family index for navigating a field in which closely related models often combine several mechanisms. Table 8 focuses on representative models whose complexity profiles and explanation mechanisms are sufficiently documented for side-by-side discussion, while Table 9 summarizes sequence- and variate-dimension scaling, memory behavior, explanation mode, overhead, and a typical caveat for each family. Figure 6 complements this discrete index with a qualitative two-dimensional placement of representative models and the main architectural lineages connecting them. Together, the figure and tables provide an entry point to the more detailed family-specific discussion below.

The advent of the Transformer architecture Vaswani et al. (2017) marked a turning point in sequence modeling, although its direct application to long-term time series forecasting was constrained by the $O(L^2)$ computational and memory cost of self-attention. Long lookback windows are often needed to capture seasonal structure and delayed effects, so reducing this bottleneck became an early priority. The subsequent evolution of DL-MTS models can be viewed as a multi-dimensional effort to improve scalability across both sequence length L and variate count N, while exposing different forms of temporal, variate, and pattern-level structure. Some approaches sparsify or restructure attention, others move computation into frequency or decomposition domains, and still others question whether attention is needed at all. This section traces that evolution from efficient Transformer variants to frequency-domain and convolutional designs, simplified linear backbones, and state-space models. Figure 5

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Table 8 Representative Contemporary Deep Learning Models for MTS Forecasting

Model	Year	L-Comp.	N-Comp.	Params	Scope	Core Innovation	Interpretability Mechanism
Informer Zhou et al. (2020)	2021	$O(L \log L)$	$O(N)$	∼11M	T	ProbSparse attention	Distillation; sparse attn maps
Autoformer Wu et al. (2021)	2021	$O(L \log L)$	$O(N)$	∼10M	T, P	Auto-Correlation; decomp	Trend-seasonal decomp viz
FEDformer Zhou et al. (2022)	2022	$O(L \log L)$[a]	$O(N)$	∼20M	T, P	Freq Enhanced Block	Sparse freq-mode attn; physical decomp
TimesNet Wu et al. (2023)	2023	$O(L \log L)$	$O(N)$	∼0.6M	T, P	2D-Variation modeling	2D heatmap viz
FNF+DBD Xu et al. (2025)	2025	$O(L \log L)$	$O(N \log N)$	∼5M	T, P	Fourier Neural Filter	Unified time-freq backbone; sparse freq viz
State Space Models & Hybrids							
S-Mamba Wang et al. (2024)	2024	$O(L)$	$O(N)$	∼10M	(V)[b]	Bi-Mamba for var corr	Selective propagation
TimePro Ma et al. (2025)	2025	$O(L)$	$O(N)$	∼1.5M	T, V	Variable-aware hyper-states	Focus Rate; Memory Entropy
Linear & Channel-Independent (CI) Backbones							
DLinear Zeng et al. (2023)	2023	$O(L)$	$O(1)$[c]	∼0.3M	T, P	Trend-remainder decomp	Linear weights; explicit decomp
FITS Xu et al. (2024)	2024	$O(L \log L)$	$O(1)$[c]	∼0.01M	T, P	Freq-domain interpolation	Amplitude scaling; phase shift
PatchTST Nie et al. (2023)	2023	$O(L)$[d]	$O(1)$[c]	∼0.5M	T	Channel-independent patching	Patch-wise temporal weights
Advanced Cross-Variate (CD) Architectures							
iTransformer Liu et al. (2024)	2024	$O(L)$	$O(N^2)$	∼5M	T, V	Inverted variate tokens	Cross-channel attn candidates
Crossformer Zhang and Yan (2023)	2023	$O(L)$	$O(N^2)$	∼2M	T, V	Two-stage attention	Variate-dimension router maps

[a] FEDformer's theoretical complexity is $O(L \log L)$ due to FFT; however, sparse mode selection significantly reduces memory.
[b] S-Mamba selective state dynamics are opaque and lack a dedicated explanation interface (Level 1).
[c] Channel-independent (CI) models provide within-variate explanations; cross-variate attribution requires an additional mechanism.
[d] PatchTST complexity is $O(P^2)$ where patch count $P \ll L$, providing near-linear scalability in practice.
[†] Temporal attribution for CI models reflects per-channel saliency only.
Scope: **T**=Temporal, **V**=Variate, **P**=Pattern.

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Table 9 Typical explanation-pillar support by architecture family. T, V, and P denote Temporal Importance, Variate Importance, and Pattern Decomposition. "Yes" indicates that the family commonly exposes a relevant internal mechanism; "Cond." indicates model- or method-dependent support; and "No" indicates that an additional attribution method is needed.

Family	Representative examples	T	V	P	Main caveat
Sparse or patched Transformers	Informer, PatchTST, Powerformer	Yes	Cond.	Cond.	Attention or patch saliency can be linked to forecast behavior through perturbation tests; channel-independent variants focus on within-channel temporal structure.
Linear and channel-independent backbones	DLinear, FITS, OLinear	Yes	No	Cond.	Temporal weights or components are accessible, but independent channels omit cross-variate influence and may limit nonlinear capacity.
Frequency and decomposition models	FEDformer, N-BEATS, N-HiTS	Cond.	Cond.	Yes	Spectral or basis components are structured, but their relation to individual variables, horizons, or causal effects is model-dependent.
Cross-variate attention and graph models	iTransformer, Crossformer, CrossGNN	Cond.	Yes	Cond.	Cross-channel scores or graph edges support relation-level inspection; behavioral and causal claims call for corresponding validation.
State-space and selective-scan models	S4, S-Mamba, TimePro	Cond.	Cond.	Cond.	States, gates, kernels, and scan dynamics can be probed, but mature temporal and variate attribution methods remain limited.
CNN, TCN, and multi-scale hybrids	ModernTCN, SCINet, TimeMixer	Yes	Cond.	Yes	Receptive fields and scale paths are inspectable, although routing and cross-scale contributions are often only qualitatively assessed.
Foundation, probabilistic, and causal extensions	Chronos, TimeGPT, CUTS	Cond.	Cond.	Cond.	Support depends on the extension: language rationales may be ungrounded, uncertainty requires calibration, and causal claims require strong assumptions.

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groups the landscape into five umbrella families, while Figure 6 places representative paradigms in the efficiency–interpretability space and traces their main architectural lineages.

Tokenization is central to both computational scaling and explanatory granularity. Time-point tokens refer to temporal positions, variate tokens expose cross-channel relations, patches summarize local intervals, and frequency or decomposition tokens represent transformed patterns. Thus, an attention matrix in Informer, PatchTST, and iTransformer refers to different units: temporal positions, within-channel patches, and variate embeddings, respectively. Mapping these signals back to timestamps or variables and reporting the aggregation across heads and layers clarifies their meaning; behavioral tests can then connect them to forecast sensitivity.

The remainder of this section discusses the umbrella families in Figure 5 and the finer-grained paradigms represented in Figure 6.

4.1 Breaking the Quadratic Barrier: The Rise of Efficient Transformers

The initial wave of innovation focused on modifying the self-attention mechanism to reduce its complexity while retaining its capacity to model long-range dependencies. Informer Zhou et al. (2020) introduced the *ProbSparse* self-attention mechanism, grounded in the empirical observation that attention scores in time series often follow a long-tail distribution. Instead of evaluating every query with equal effort, it selects the top-u queries expected to carry the most informative attention patterns, reducing complexity to $O(L \log L)$. A self-attention distilling operation further compresses the feature representation between layers, allowing progressively coarser processing of long inputs. These choices made Informer an influential early route for scaling Transformer architectures to LSTF tasks. They also change what an attention visualization contains: the sparse map represents aggregation over selected queries and is more difficult to map back to all historical positions. Perturbation and stability tests can establish how closely this pattern tracks temporal influence Bastings and Filippova (2020).

Building on this efficiency-driven momentum, subsequent models sought to refine the attention mechanism further. Wu et al. (2021) proposed a significant architectural shift by incorporating a series decomposition block as an internal model component. Repeated decomposition progressively separates trend and seasonal components instead of postponing this structure to preprocessing or a final diagnostic. More critically, Autoformer replaces self-attention with an Auto-Correlation mechanism that discovers period-based dependencies at the series level and aggregates similar subseries. By leveraging the Fast Fourier Transform (FFT), it also achieves $O(L \log L)$ complexity. The learned lags and decomposition components provide a more structured view than an undifferentiated attention matrix: users can inspect which periodic relationships are aggregated and how trend and residual components contribute to the forecast. This transparency remains tied to the decomposition and periodicity assumed by the architecture, but it is a substantive part of the forecasting path rather than a separate post-hoc layer.

A parallel line of inquiry questioned the conventional approach of embedding all variables observed at one time step into a single token. Zhang and Yan (2023) argued

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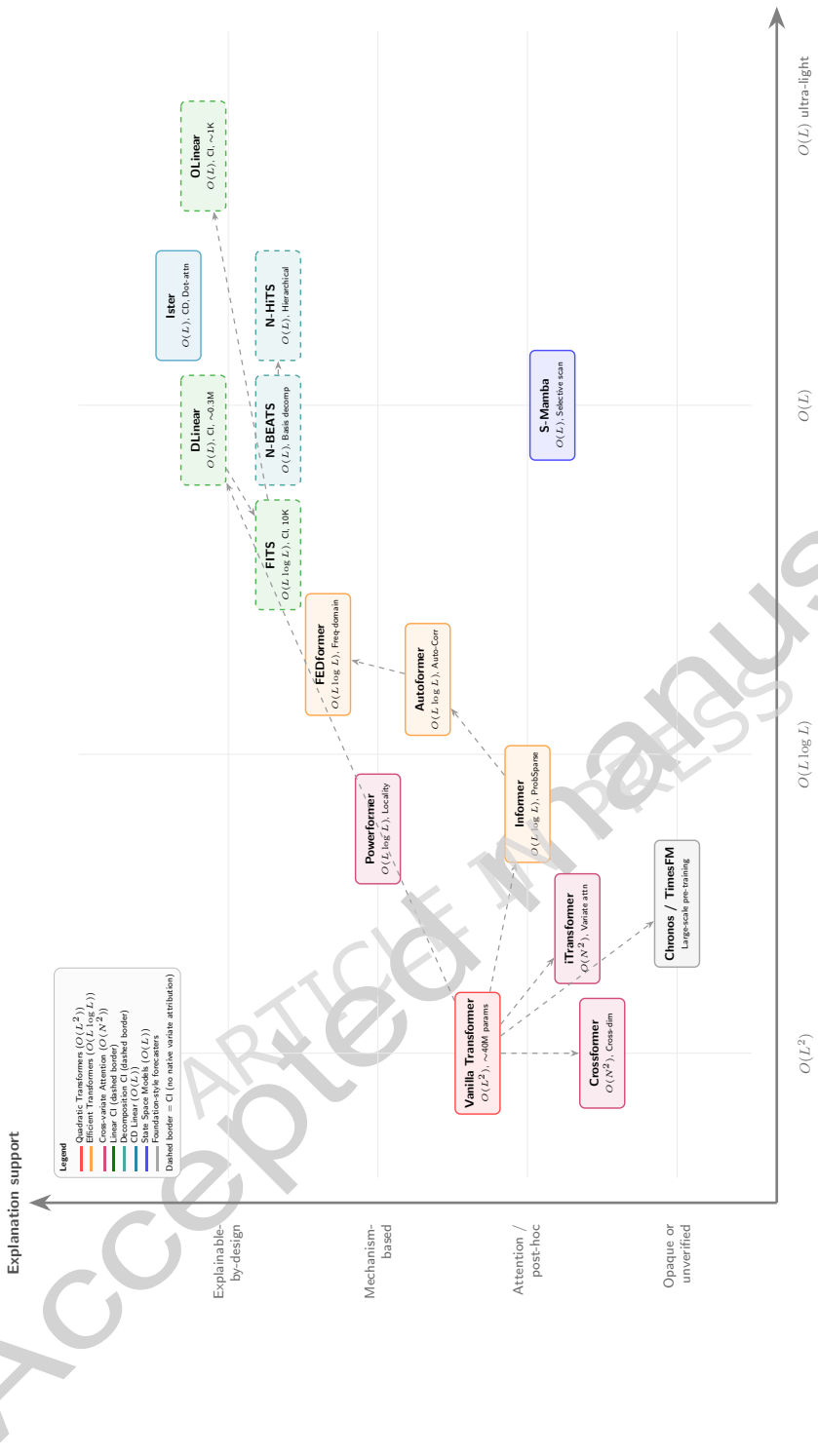


Fig. 6 Two-dimensional classification of representative MTS forecasting models by computational efficiency (horizontal) and interpretability level (vertical, Levels 0–3 per Section 3). Model positions are qualitative. Dashed borders denote channel-independent (CI) architectures rated Level 3†, whose architecture-internal explanation scope covers Temporal Importance and Pattern Decomposition within each variate (see Section 3). Arrows trace major architectural lineages. Ister is detailed in Section 5.

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that this representation makes cross-dimension dependencies difficult to model explicitly. Its Dimension-Segment-Wise (DSW) embedding treats each variable-specific interval as a separate segment, and the Two-Stage Attention (TSA) mechanism first captures cross-time dependencies within a dimension before using a router to exchange information across dimensions. The router avoids evaluating every cross-dimension pair directly and maintains linear complexity with respect to the number of variables in the stated design. This separation gives the model distinct temporal and cross-variable attention objects at their respective granularities. Router interventions and attention perturbations can relate these objects to forecast influence.

An alternative approach to improving attention efficiency and interpretability is to introduce an explicit locality bias. Hegazy and Schaar (2025) proposed Powerformer, which replaces the standard non-causal attention weights with a weighted causal multi-head attention (WCMHA) mechanism. The key insight is that in many time series, pairwise temporal correlations decay following a power-law distribution, yet standard self-attention assigns uniform prior weights to all historical time steps, ignoring this natural structure. By applying power-law decay or Butterworth filtering to the causal attention connections, Powerformer injects a locality inductive bias that better reflects the temporal correlation structure of real-world data. Combined with the patching strategy from PatchTST, the source study reports strong forecasting results across 47 evaluation tasks and attention patterns shaped by the imposed locality prior. These patterns make the temporal bias inspectable, and perturbation or stability tests can relate them to model behavior. PatchTST's channel-independent design confines its attention maps to temporal importance within each channel; cross-variate ranking requires a separate mechanism.

Critical Analysis and Trade-offs: Collectively, these models reduce the quadratic attention bottleneck and make Transformer architectures more practical for LSTF. Informer emphasizes sparse query selection, Autoformer combines decomposition with periodic aggregation, Crossformer separates temporal and cross-variate interaction, and Powerformer introduces an explicit locality prior. The resulting structures can make selected model operations easier to inspect, but they also introduce assumptions about the data. ProbSparse favors a dominant subset of temporal relationships, Auto-Correlation relies on recurring periodicity, Crossformer's segment representation can blur instantaneous cross-variable effects, and Powerformer's decay prior may be less suitable when distant non-local dependencies dominate. Their forecasting and explanatory benefits therefore vary with the structure of the series.

4.2 The Frequency Domain Paradigm: A New Lens for Interpretability

As efficient Transformers matured, a new paradigm shifted attention from the time domain to frequency-domain analysis. Drawing on classical signal processing, these models represent temporal structure through constituent frequencies and learned decompositions, which can be more economical when recurring patterns dominate the signal. Zhou et al. (2022) introduced a Frequency Enhanced Block (FEB) and Frequency Enhanced Attention (FEA) operating through Fourier and Wavelet transforms. After each FFT, the model retains a fixed, sparse set of M Fourier modes

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rather than mixing the entire spectrum. This reduces the practical cost and memory footprint of frequency mixing while preserving the $O(L \log L)$ complexity of the FFT/IFFT backbone. The selected modes, amplitudes, phases, and decomposition components offer an inspectable global account of seasonalities and trends that differs from point-wise temporal attention. The accompanying Kolmogorov–Smirnov analysis characterizes the input and forecast distributions in the source study, while component-level behavioral tests address explanation faithfulness.

Extending this representation-oriented view, Wu et al. (2023) converts a one-dimensional time series into a collection of two-dimensional tensors. FFT is first used to identify prominent periodicities, after which the series is reshaped into panels whose axes describe intra-period and inter-period variation. This transformation allows image-oriented backbones such as Inception blocks to capture local variation within a cycle and longer changes across cycles in the same representation. The resulting panels and heatmaps provide intuitive views of how structures evolve at both scales and can be easier to relate to periodic phenomena than a high-dimensional hidden state. Component ablation and perturbation can connect these views to the behavior of an individual forecast.

More recently, Xu et al. (2025) proposed the Fourier Neural Filter (FNF), which combines local time-domain and global frequency-domain processing in a unified backbone and uses Fourier operations for $O(N \log N)$ channel mixing. Unlike designs that add a spectral module to a general-purpose backbone, FNF treats the temporal and channel-frequency views as complementary parts of the core architecture. Its Dual Branch Design (DBD) is motivated through an information-bottleneck analysis of information flow and representation capacity. The source study reports results across 11 public benchmarks spanning energy, meteorology, transportation, environmental, and natural domains, suggesting that frequency-aware inductive biases can provide an effective alternative to general-purpose sequence backbones. We interpret these numerical comparisons within the datasets, preprocessing, and tuning conditions reported by that study. Parallel to these spectral innovations, *ModernTCN* Luo and Wang (2024) revisits convolutional forecasting through large-kernel depth-wise convolutions, patching, and multi-scale feature extraction. Large kernels provide a broad temporal receptive field without constructing a full pairwise attention matrix, while depth-wise processing keeps the main temporal operation separate for each channel and gives the architecture $O(L \cdot N)$ scaling in the sequence and variate dimensions. The learned convolutional filters and their temporal responses can be visualized within each channel, making recurring local patterns easier to inspect than an undifferentiated hidden representation. This supports Temporal Importance at the channel level; cross-variate influence is carried by subsequent channel-mixing components and can be evaluated there.

Critical Analysis and Trade-offs: Frequency-domain models offer a structured alternative to attention-based forecasting, often with quasi-linear $O(L \log L)$ complexity and components that can be related to daily, weekly, or other recurring cycles. This connection can be especially useful when domain experts already reason in terms of periodicity and decomposition. Its value, however, depends on the selected basis and on the presence of periodic or quasi-periodic structure. Global Fourier coefficients

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may not identify the local event that activated a component, finite windows introduce leakage, and abrupt or aperiodic events may be poorly represented by a fixed basis. Inverse reconstruction, time-localized sensitivity, and component ablation can help determine whether an interpretable spectral component materially affects the forecast.

4.3 Radical Simplification: The Linear Renaissance

While SSMs pursue linear efficiency through complex recurrent dynamics, the *Linear Renaissance* achieves the same $O(L)$ goal through radical architectural simplification. The pursuit of architectural complexity was challenged by Zeng et al. (2023), who compared Transformer forecasters with the simple linear model DLinear. In the long-term forecasting settings examined in that study, DLinear outperformed several Transformer baselines, showing that simple temporal mappings can capture the relevant dynamics in suitable settings. The result also reinforced the value of a strong simple baseline: improvements attributed to a new sequence mechanism can be difficult to interpret if temporal order and basic decomposition have not first been modeled adequately. DLinear separates the input into trend and remainder components and learns a linear projection for each, preserving the ordering of the lookback window while maintaining $O(L)$ complexity. Its weights and decomposition components provide a direct view of within-channel temporal structure and can be tested by perturbing selected lags or components. Its channel-independent architecture gives the explanation a within-variate scope, as in FITS and PatchTST.

This "renaissance of simplicity" echoed earlier work like N-BEATS Oreshkin et al. (2019), a deep learning model that, while not strictly linear, eschewed complex recurrent or attention-based components. N-BEATS uses a deep stack of fully-connected layers with a basis expansion principle to explicitly model and forecast trend and seasonality components. Its hierarchical decomposition provides clear, interpretable outputs without sacrificing performance, demonstrating that carefully designed deep architectures could be both powerful and transparent.

More recently, FITS Xu et al. (2024) reframed forecasting as interpolation in the frequency domain. It transforms each channel, retains a low-frequency representation, and applies a single complex-valued linear layer before reconstructing the forecast. The reported configurations use roughly 10,000 parameters while retaining $O(L \log L)$ FFT complexity, illustrating how a compact model can exploit the concentration of many time-series signals in lower frequencies. The learned amplitude scaling and phase shifting provide a direct view of how dominant frequencies are extrapolated and can be related to changes in periodic magnitude or timing. This interpretation applies within each independently processed variate; reporting truncation and normalization choices clarifies which spectral information is available to the model.

The resurgence of simplified models has also produced alternatives to complex attention. FreTS Zhou et al. (2024) applies a linear layer to spectral features, using the frequency domain as a compact space for learning dependencies. TiDE Das et al. (2023) follows a different route by combining an MLP encoder with a residual linear decoder and explicit static and dynamic covariates. Its $O(L + N)$ scaling avoids pairwise attention, while the separated covariate pathways make it possible to inspect

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how known external inputs enter the forecast. Behavioral analysis can further connect these pathways to local predictions. Together with FITS and DLinear, these designs show how spectral decomposition, explicit covariates, and compact backbones can remain competitive across suitable forecasting settings.

Recent linear and simplified architectures also explore ways to expose internal structure without adding a separate post-hoc explainer. The Concept-enhanced Physics-informed Transformer (CPformer), for example, integrates a concept-based module with five domain-agnostic concepts, creating inspectable concept-level objects within a Transformer backbone Ma et al. (2025). OLinear uses fixed orthogonal transforms such as the Discrete Cosine Transform (DCT) and reports competitive results with roughly one thousand parameters in its source configurations Yue et al. (2025). These designs illustrate how efficiency and selected forms of transparency can be co-designed, with task-specific evaluation establishing the faithfulness and domain meaning of their internal objects. Section 4.4 examines the different transparency challenges that arise for state-space models.

Critical Analysis and Trade-offs: Linear and frequency-based models invite a reassessment of when additional architectural complexity is useful. They can combine low cost, inspectable components, and competitive forecasting performance, but their capacity may be limited for strongly nonlinear or non-stationary dynamics. The channel-independent design that enables efficient per-channel processing gives these models a clear temporal and pattern scope, represented by the Level $3^{\dagger}$ designation. Their weights usually describe a global temporal mapping; normalization checks, basis analysis, and perturbations of highlighted lags or components can connect that mapping to an individual forecast.

4.4 State Space Models: A Linear-Complexity Alternative to Attention

State Space Models (SSMs) have become a prominent efficient paradigm for modeling long sequences and an alternative to Transformer architectures. Rooted in classical control theory, SSMs map a 1D input sequence $u(t)$ to an output $y(t)$ through a latent state vector $x(t)$. They can be formulated as either a parallel scan or a convolution, enabling linear complexity of $O(L)$ with respect to sequence length L and avoiding the quadratic bottleneck of full self-attention.

The S4 model helped establish the modern Structured State Space Model paradigm Gu et al. (2022). It uses the HiPPO (High-order Polynomial Projection Operator) framework to initialize state matrices and a low-rank parameterization that permits stable diagonalization, reducing the computation to Cauchy-kernel operations. S4 was the first model evaluated in its study to solve the Path-X task in the Long Range Arena at sequence length 16,384 and reported up to 30× faster computation and 400× lower memory use than the linear state-space baselines considered there. These figures apply to the source study's Path-X setup and evaluated baselines.

Building upon S4's foundations, the Mamba architecture introduced a selective SSM (S6), in which key state-space parameters become functions of the current input Wang et al. (2024). This allows the model to retain, propagate, or suppress information

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according to content rather than applying the same transition dynamics at every position. An implementation-aware selective scan preserves linear sequence scaling while supporting parallel training, giving Mamba a practical route to long-context modeling without constructing an attention matrix. Selectivity addresses an important limitation of fixed SSM dynamics, but it also changes the object available for explanation: information is distributed across input-dependent gates and evolving latent states rather than explicit pairwise attention weights.

Adapting this paradigm to multivariate forecasting, Wang et al. (2024) proposed S-Mamba. Similar to the inverted strategy of iTransformer, it represents variables as tokens and uses a Bi-directional Mamba Variable-wise Coder (Bi-Mamba VC) to model their relationships through a linear scan rather than full quadratic attention. Temporal information within each variate is then projected through a feed-forward component. This design preserves the sequence-processing advantages of Mamba while providing an explicit route for cross-variate interaction; its realized cost still depends on lookback length, hidden width, variate count, and implementation.

S-Mamba reports particularly strong results on high-dimensional, periodic datasets Wang et al. (2024), including Traffic (MSE 0.382) and Electricity (MSE 0.139) in the study's evaluation. On PEMS07, it uses 1.47GB of GPU memory compared with 2.38GB for the evaluated iTransformer configuration. Its advantage is less pronounced on lower-dimensional, non-periodic datasets, where simpler linear models remain competitive, suggesting that the benefit depends on the structure of the data.

Extending the SSM paradigm to multi-latency dependencies, Ma et al. (2025) introduced TimePro, which constructs variable- and time-aware hyper-states to retain both cross-variable context and salient temporal information. Conventional variable-token designs can compress an entire lookback window into a single vector before exchanging information across channels; TimePro instead preserves finer temporal states for each variate and uses focal time-point selection to represent relationships that operate at different delays. This is useful when one variable responds immediately while another carries a delayed effect. The source study reports competitive results on eight real-world benchmarks together with low parameter, FLOP, and memory requirements. The focal-point selections and variable-specific states also provide natural diagnostic targets, and interventions on these states can test their influence on the forecast.

A complementary research direction uses specialized pre-training to make selected Mamba dynamics easier to inspect. Yan et al. (2025) developed Repetitive Contrastive Learning (RCL), a token-level contrastive framework that improves temporal selectivity without adding inference overhead. RCL introduces two quantitative summaries: *Focus Rate*, which measures the proportion of selective attention allocated to recent and distant tokens, and *Memory Entropy*, which measures the concentration of state-memory allocation across time. These metrics provide a useful starting point for studying how SSMs retain temporal information and for developing finer temporal attribution methods.

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Hybrid SSM-Transformer Architectures.

A further direction combines the linear-complexity processing of SSMs with the local flexibility of attention Patro et al. (2024). The State Space Transformer (SST), for example, routes long-range dependencies to Mamba blocks and short-range variation to Local-Window Transformers. This division of labor reflects a useful architectural intuition: recurrent state updates can summarize long histories economically, while local attention can revisit transient events at finer resolution. It may therefore offer a better balance than assigning both roles to a single operator. At the same time, the routing mechanism becomes part of the model's reasoning path. Understanding a forecast may require examining which patterns were sent to each branch, whether those allocations are stable for nearby inputs, and how the output changes when one branch is suppressed. Router diagnostics and per-expert attribution are natural extensions of the hybrid design.

Foundation-Scale SSMs.

The success of foundation models in NLP and vision has inspired efforts to scale SSMs for time series Patro et al. (2024). TSMamba represents an early foray into this direction, demonstrating transfer learning and zero-shot forecasting capabilities with linear-time scaling. By pre-training on large, diverse time series corpora, TSMamba can generalize to unseen domains without fine-tuning, addressing a key limitation of task-specific models. The model's cross-channel attention compression mechanism reduces the quadratic variate-dimension cost while maintaining multivariate modeling capacity. From an interpretability perspective, foundation-scale SSMs raise questions about how explanations should be provided for zero-shot predictions on novel domains where no training-time provenance exists.

Critical Analysis and Trade-offs: The progression from S4 to Mamba, S-Mamba, and hybrid or foundation-scale variants establishes a linear-time alternative to quadratic attention for long-range sequence modeling. The main trade-off shifts from computational cost toward the interpretation of latent state dynamics. Selective gates, kernels, memory decay, and input-dependent transitions are less directly connected to named temporal, variate, or pattern contributions than the components of decomposition models. Impulse-response analysis can reveal retained timescales; gate interventions can test which inputs alter memory selection; state or kernel ablation can connect latent pathways to forecast changes; and stability analysis can examine whether these signals persist across nearby inputs and checkpoints. Control-theoretic observability remains useful for understanding whether states can be recovered from outputs, but explaining an individual forecast still requires linking these diagnostics to input-level behavior.

4.5 Architectural Diversification and Hybrid Models

Beyond efficient attention, frequency analysis, linear simplification, and state-space modeling, the field has developed a broad range of hybrid designs and specialized components for representation learning, multi-scale dynamics, and cross-variable dependency modeling. This diversification reflects a growing recognition that no single architecture is best suited to every dataset or deployment requirement.

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4.5.1 Rethinking Attention: Inverted and Adaptive Designs

A significant line of research has re-imagined the Transformer's core representation. The Inverted Transformer, or iTransformer, treats the full history of each individual time series as one token and applies attention across variables, scaling as $O(N^2)$ in the variate dimension Liu et al. (2024). Its feed-forward layers learn temporal representations within each variate, while cross-token attention models relationships among variates. This inversion has proved effective for multivariate forecasting and makes the channel-to-channel attention matrix directly available for inspection. The matrix aligns naturally with a variate-level question, while behavioral tests determine how its aggregation weights relate to predictive influence.

CARD extends this channel-centric view by alternating attention along the token and channel dimensions and introducing a signal-decay loss to regularize temporal dynamics Wang et al. (2024). The alternating stages separate two kinds of interaction that are entangled in a single flattened representation, giving analysts distinct temporal and cross-channel objects to examine. Other models combine deep sequence processing with established time-series operators. ETSformer integrates exponential smoothing and frequency components into a Transformer architecture, so level, growth, and seasonal effects remain visible within the forecasting path Woo et al. (2022). Temporal Convolutional Networks provide another important point of comparison: causal and dilated convolutions expand the receptive field while preserving temporal order and have long served as an effective alternative to recurrent sequence models Bai et al. (2018).

These inverted and adaptive attention designs address a fundamental limitation of both conventional Transformers and CI linear models: the former entangle cross-variate structure in a shared embedding, while the latter process each variate independently. By explicitly applying attention across the variate dimension, iTransformer and CARD expose cross-channel interaction signals for Variate Importance analysis, with $O(N^2)$ scaling that becomes costly for high-dimensional datasets.

The contrast between iTransformer and DLinear illustrates this complementarity through the three-pillar lens. DLinear ($O(L)$, ∼0.3M parameters, CI) provides inspectable per-channel temporal weights ($T^{\dagger}$) and explicit trend–remainder decomposition (P). iTransformer ($O(N^2)$, ∼5M parameters, CD) adds cross-channel attention maps for validation-backed Variate Importance analysis (V), while offering less explicit pattern decomposition. DLinear is therefore attractive for low-cost temporal or pattern inspection, whereas iTransformer addresses applications in which cross-variate analysis is central.

4.5.2 Multi-Scale and Adaptive Mechanisms

A parallel trend involves the development of multi-scale mechanisms that allow models to dynamically adjust to varying temporal patterns. TimeMixer and its successor, TimeMixer++, exemplify this by designing a multi-scale decomposition framework that mixes past-future information across diverse resolutions Wang et al. (2024a, 2025). Beyond enhancing Pattern Decomposition (our third pillar), this hierarchical approach implicitly improves efficiency by processing global trends at lower sampling

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rates, thereby reducing the computational redundancy inherent in single-scale models. Pathformer introduces an adaptive multi-scale path selection mechanism that unifies the resolution and distance dimensions, allowing the model to learn the optimal pathway for information flow Chen et al. (2024). This principle of adaptivity is explored through various lenses: AMD provides a framework for adaptive multi-scale decomposition Hu et al. (2025), LIFT focuses on dynamically modeling lead-lag relationships between series Zhao and Shen (2024), and FAN uses frequency-adaptive normalization to improve representation learning Ye et al. (2024). Some models approach adaptivity through clustering; Chen et al. (2024) proposed a channel clustering module to group similar variables, while Duet employs a dual-clustering framework to enhance modeling Qiu et al. (2025). Further refinements come from specialized filtering and attention mechanisms, such as the patch-level spatio-temporal graph filtering in TimeFilter Hu et al. (2025) and the enhanced attention mechanisms designed to resolve low-rank bottleneck issues in Transformers Yue et al. (2025); Wang et al. (2025).

The multi-scale paradigm offers a nuanced form of Pattern Decomposition by learning to separate signals at different temporal resolutions. A low-resolution branch may emphasize slowly varying trends, whereas a high-resolution branch retains short fluctuations and local events; inspecting their outputs can therefore reveal which temporal scales are active for a forecast. Because several branches may encode overlapping structure, domain labels are most informative when supported by branch removal, scale-specific perturbation, or routing-stability tests. These analyses can make the routing decisions in Pathformer and the scale mixing in TimeMixer easier to relate to forecast changes.

4.5.3 Advanced Frequency-Domain and Convolutional Architectures

The frequency domain continues to be a fertile ground for innovation. Fredformer explicitly addresses frequency bias by patching series in sub-frequency bands Piao et al. (2024), while WaveForM enables end-to-end learning directly in the wavelet domain Yang et al. (2023). Others introduce novel representations, such as FRNet, which uses complex-valued rotation networks to model dynamic periodicities Zhang et al. (2024). A key challenge in spectral methods is correcting for biases introduced by data processing; FreDF tackles this by proposing a frequency-domain MAE loss to counteract auto-correlation biases from data differencing Wang et al. (2025). Efforts to create more powerful representations have led to dynamic frequency fusion techniques coupled with independent transfer functions Zhang et al. (2024) and frameworks for periodic decomposition Dai et al. (2024a).

Concurrently, recurrent and convolutional designs have renewed alternatives to attention. SegRNN uses segmental iteration to process long sequences Lin et al. (2023), while PGN proposes a parallel gating network Jia et al. (2024). SCINet employs recursive down-sampling and an interactive learning tree to capture multi-resolution features Liu et al. (2022), and LightCTS uses grouped convolutions for efficient temporal modeling Lai et al. (2023). TimeBase achieves compact forecasting through basis decomposition Huang et al. (2025), while TiDE uses a global residual MLP over covariates Das et al. (2023). These approaches broaden the design space of efficient backbones and provide different forms of temporal or pattern structure.

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These frequency-domain and convolutional approaches share a common strength: they provide structured Pattern Decomposition through physically meaningful spectral representations or hierarchical feature extraction. Their channel-independent or channel-concatenated implementations primarily support temporal and pattern analysis, while variate attribution depends on the accompanying channel-mixing mechanism.

4.5.4 Graph-based Cross-Variate Modeling

Graph Neural Networks (GNNs) offer a principled way to model cross-variate dependencies when relational structure is central. Instead of allowing every variable to interact through an unstructured dense layer, they represent channels as nodes and use edges to control how information is exchanged. StemGNN combines graph spectral analysis with temporal spectral modeling, allowing dependencies among series to be learned jointly in the frequency domain Cao et al. (2020). CrossGNN uses separate graph modules for cross-scale and cross-variable relations Huang et al. (2023); TPGNN learns a time-varying polynomial graph to accommodate evolving interactions Liu et al. (2022); and LGI infers a latent graph directly from data when no physical topology is available Satorras et al. (2022). A known sensor network can make the resulting pathways easy to relate to the application, while a learned graph can reveal recurring dependencies that were not specified in advance. In either case, an edge describes modeled dependence rather than establishing a causal effect.

From an interpretability standpoint, GNN-based models occupy a distinctive position: the learned or pre-defined graph supplies an explicit representation of cross-variable information flow that is absent from channel-independent linear models. Edge weights, neighborhoods, and time-varying adjacency patterns can all be inspected, and node or edge masking can test whether a highlighted relation materially affects a forecast. This advantage comes with $O(N^2)$ complexity when a dense graph must be constructed and with the risk that learned edges reflect correlation, common causes, or dataset artifacts. The issue is especially important in non-stationary settings, where a graph that appears plausible on average may fail to describe the relationships active for a particular forecast. Sparse graph constraints, stability analysis, and domain validation are therefore useful complements to visualization.

5 Emerging Paradigms in the Efficiency-Explainability Landscape

The architectural developments reviewed in Section 4 have considerably expanded the design space of MTS forecasting, yet the relationship between efficiency and explainability remains open. Emerging paradigms approach this question from several directions: decomposition models expose named temporal components, causality-aware models seek intervention-relevant structure, adaptive mechanisms make selected forms of distribution shift explicit, and foundation models exchange task-specific training for large-scale pre-training and transfer. Figure 7 positions these directions qualitatively within the efficiency–explainability space. We examine how each

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paradigm changes the source of computational cost, the form of explanation available, and the assumptions under which that explanation is meaningful.

5.1 Intrinsically Interpretable Architectures by Design

A primary route to intrinsic interpretability is to decompose a series into meaningful components such as trend and seasonality. N-BEATS uses fully connected blocks with a doubly residual topology and constrained basis expansions Oreshkin et al. (2019). Each block produces a backcast that removes the part of the input already represented and a forecast that contributes to the final sum, allowing later blocks to focus on the remaining structure. In its interpretable configuration, polynomial and Fourier bases make the trend and seasonal contributions directly inspectable, and the component forecasts can be plotted separately before they are added. This organization is useful because the explanation follows the same additive path as the prediction rather than being generated by a separate explainer. The interpretable configuration performs comparably to its unconstrained counterpart on the benchmarks examined in the original study. Although N-BEATS was developed for univariate datasets, its basis expansion and residual decomposition have influenced multivariate implementations and successors such as N-HiTS Challu et al. (2022) and Ister Cao et al. (2024). In the usual channel-independent deployment, it supports Temporal Importance and Pattern Decomposition but not native Variate Importance, which motivates the Level 3† designation.

Building on this foundation, N-HiTS introduces multi-rate sampling and hierarchical interpolation for long-horizon forecasting Challu et al. (2022). Different max-pooling rates encourage early stacks to represent smoother, low-frequency behavior and later stacks to retain increasingly fine temporal detail. Each block predicts a compact coefficient set that is interpolated to the full horizon, so a long output does not require an equally large dense forecast head. This design reduces parameter and computational cost; the original study reports a 45-fold training-time speedup over the Transformer configurations evaluated in its long-horizon experiments. The hierarchy also gives readers a resolution-by-resolution view of the forecast, which can be informative when long-term direction and short-term variation play different operational roles. N-HiTS inherits the channel-independent topology of N-BEATS, so its structured temporal components do not provide native cross-variate attribution.

The hierarchy makes resolution-specific forecasts inspectable, although different stacks can partly substitute for one another and a low-frequency component need not correspond uniquely to a domain-defined trend. Basis and interpolation choices, together with component ablations, help determine whether labels such as “trend” or “seasonality” accurately describe how the fitted model forms a forecast.

Beyond basis expansion, recent work has explored other structured approaches to intrinsic interpretability. Yue et al. (2025) introduced OLinear, which replaces learnable linear projections with fixed orthogonal transforms such as the DCT, DFT, or Legendre polynomials. Orthogonality separates basis directions, while the fixed transforms remove learnable parameters from the projection stage and enable configurations with approximately 1K parameters. The resulting coefficients give a direct

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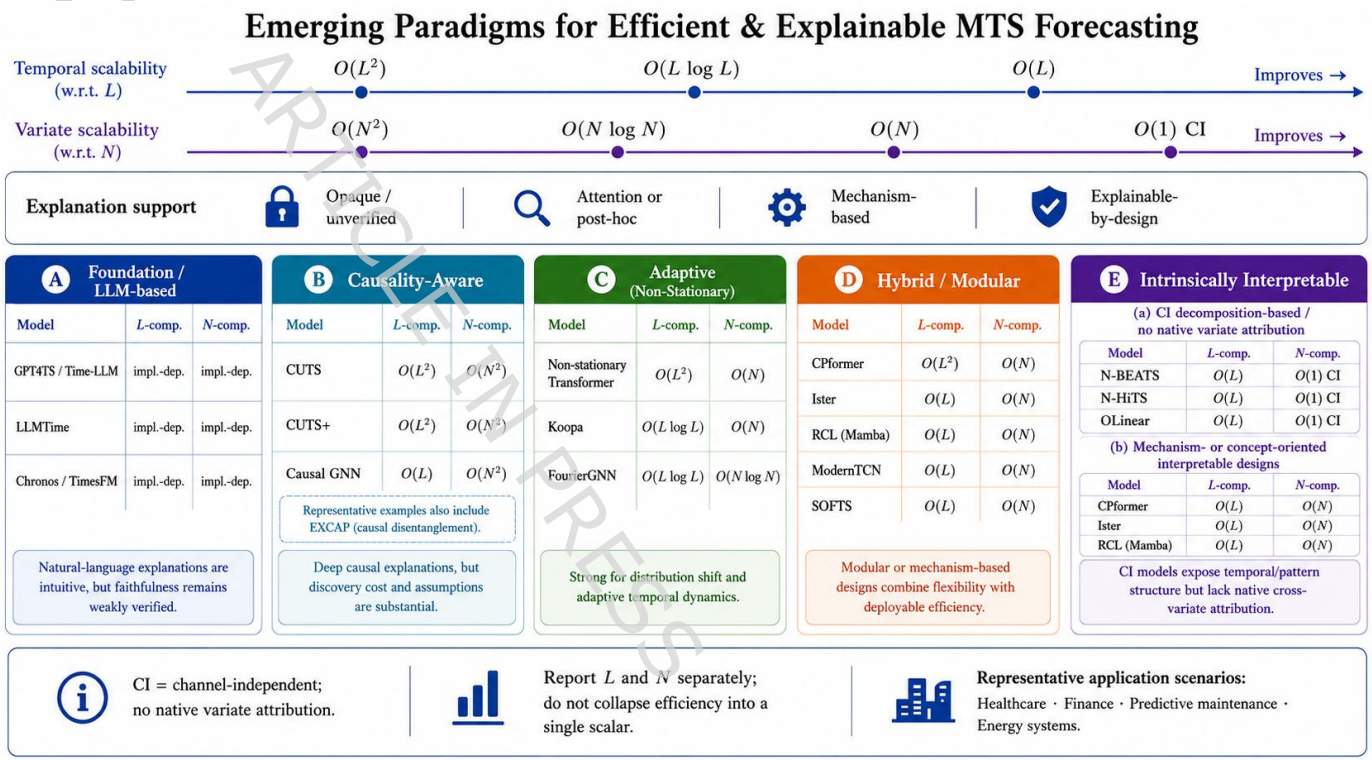


Fig. 7 Qualitative landscape of emerging MTS forecasting paradigms by computational efficiency and explanation specificity. Dashed borders indicate channel-independent models whose architecture-internal explanations focus on within-variate structure.

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view of the frequency components used by the model, and component ablation can further connect those coefficients to individual forecasts.

The interpretation is consequently tied to the selected basis and preprocessing: DCT coefficients emphasize smooth oscillatory structure, DFT coefficients retain amplitude and phase, and polynomial bases emphasize lower-order shape. Normalization, truncation, and channel-wise processing can change their scale, so coefficients from different models or variates should be compared with care.

Taking a complementary approach, Cao et al. (2024) proposed Ister, which combines hierarchical temporal decomposition with Dot-attention. It separates trend and seasonal components and further divides seasonality into several periodic subcomponents. Dot-attention reduces temporal complexity from $O(L^2)$ to $O(L)$, while the resulting channel- and period-specific components provide an inspectable account of the model's decomposition. Their use as attribution signals can be strengthened through behavioral validation.

A distinct paradigm uses *concept bottlenecks* to align intermediate representations with named, human-interpretable concepts. Instead of asking a user to interpret hundreds of latent coordinates, the model passes information through a smaller set of quantities with intended semantic meaning. van Sprang et al. (2024) uses Centered Kernel Alignment (CKA) to connect time-series Transformer representations with predefined temporal features and autoregressive proxies, enabling users to inspect concept activation and intervene on selected concepts under distribution shift. CPformer defines level, growth, periodicity, volatility, and exogenous pressure and combines a learnable bottleneck with physics-informed regularization Ma et al. (2025). These concepts provide a concise vocabulary for discussing how a forecast is formed and can support counterfactual inspection by varying one concept while holding the others fixed. In its six-dataset evaluation, CPformer reduces MSE relative to the tested FEDformer configuration by 23% on Electricity, 44% on Traffic, and 61% on ILI; removing concept alignment increases MSE by 113% in the reported ablation. These results support the contribution of the concept module in that setting. Evaluation of concept purity, intervention behavior, and agreement with domain knowledge can establish how well the five concepts transfer across domains.

Critical Analysis: A Rebalanced Trade-off?

N-BEATS, N-HiTS, OLinear, and Ister illustrate how lower computational cost can coexist with structured temporal or pattern components. Their trade-offs shift toward the choice of basis and explanatory scope. N-HiTS, for example, introduces a smoothness bias through hierarchical interpolation: synthesizing long-horizon forecasts from compact coefficients can reduce cost and suppress high-frequency noise, but may be less suitable when structural breaks or rapid volatility dominate. More generally, decomposition-based explanations are clearest for the components explicitly represented by the architecture, while cross-variate influence calls for an accompanying channel-interaction mechanism.

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5.2 Causality-Aware Forecasting

Causality-aware forecasting extends correlation-based importance analysis toward relationships that support intervention-oriented questions. As Gong et al. (2023) note, conventional deep learning models, including those with attention mechanisms, primarily capture statistical association. One major branch of this paradigm uses Granger causality, under which the past of a series X provides predictive information about a series Y beyond the past of Y itself. This principle has been integrated with Graph Neural Networks (GNNs), where a learned or predefined graph structures the pathways through which information is propagated. The resulting representation connects forecasting behavior to an explicit relational hypothesis that can be examined and tested.

A complementary approach is rooted in the Structural Causal Model (SCM) framework, which defines causal relationships through a set of functional assignments. Recent work has explored methods for discovering these SCMs from observational time series data, often involving score-based optimization in acyclic graph spaces. Once a causal graph is inferred, it can support counterfactual predictions—for instance, answering the question, "What would the energy demand have been if the factory's primary machine had not been shut down?" Such explanations are interventional in form and can align naturally with decision-making, provided that the graph and functional assumptions are supported by the application.

The inferential target determines how these relations can be interpreted. Granger-style predictability asks whether the past of one series improves forecasts of another under the observed conditioning set. Intervention effects additionally depend on graph structure, temporal order, treatment of confounding, and functional assignments. Irregular sampling, contemporaneous effects, latent common causes, and changing policies can all separate predictive edges from causal effects. Graph recovery, forecasting performance, and intervention validity are therefore complementary evaluation questions.

Recognizing limitations of existing time-series XAI methods, EXCAP Wang et al. (2025) organizes explanation quality around temporal coherence, pattern-centric explanation, causal disentanglement, and faithfulness to model behavior. It combines an attention-based segmenter, a causal-structure decoder guided by a pre-trained graph, and latent aggregation for representation stability. Its theoretical and perturbation-based analyses of smoothness and stability complement the three explanatory questions used in this survey, while the causal interpretation still depends on the assumptions of the guiding graph.

Critical Analysis: The Price of Causal Fidelity

The pursuit of causal explanations introduces trade-offs between identification assumptions and computational feasibility. As detailed by Gong et al. (2023), causal discovery from observational data commonly relies on causal sufficiency, acyclicity, stationarity, or related assumptions that may be difficult to verify in practice. Constraint-based methods can scale exponentially with the number of variables, while score-based methods search a super-exponential graph space, so the cost of discovering

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or updating a graph may offset efficiency gains in the downstream forecaster. Evaluation can connect graph recovery, forecasting accuracy, and intervention validity as distinct outcomes. Sensitivity to lag choice, hidden confounders, sampling frequency, and graph regularization is particularly important when the learned structure supports decisions.

5.3 Adaptive Mechanisms for Non-Stationary Series

Non-stationarity—where the statistical properties of a time series, such as mean and variance, change over time—is a pervasive challenge in real-world forecasting. Standard models, particularly Transformers, often implicitly assume stationarity, leading to performance degradation when faced with distributional shifts. Liu et al. (2022) identified this as an "over-stationarization" problem, where standard normalization techniques efface the distinct statistical signatures of individual series, causing the self-attention mechanism to produce near-uniform weights. To counteract this, they proposed a paradigm that explicitly models and adapts to non-stationarity. Their framework consists of two key modules: *Series Stationarization* and *De-stationary Attention*. The first module normalizes each time series instance independently to a zero mean and unit variance, feeding this temporarily stationary version into the model. The crucial innovation lies in the De-stationary Attention module, which reinjects the original series' statistical properties (mean and variance) back into the attention calculation. This allows the model to compute attention scores that are aware of the original scale and dynamics of each series, producing instance-specific attention patterns. The original statistics are then restored to the final forecast via a de-normalization step, ensuring the output is correctly scaled.

Preventing over-stationarization reduces MSE by up to 49% relative to the standard Transformer configurations evaluated in the original study Liu et al. (2022). The architecture also makes the normalization and reinjection of instance statistics explicit, providing an inspectable account of how level and scale affect attention. Perturbation and stability analysis can connect the resulting attention maps to forecast behavior alongside predictive accuracy.

Critical Analysis: Scope of the Adaptation.

The adaptive mechanism proposed by Liu et al. (2022) targets variation in selected statistical moments. Its effectiveness therefore depends on how well mean and variance normalization represents the shift of interest. This treatment addresses changes in level and scale, while changing seasonality, evolving volatility, and abrupt structural breaks call for additional mechanisms. For example, a shift in seasonal period from 12 months to 10 months lies outside a representation focused on mean and variance. Broader adaptation can combine moment correction with structural-change detection while retaining measurable efficiency and explanation properties.

5.4 Interpretability in High-Stakes Applications

The emerging paradigms discussed above are particularly relevant to healthcare, industrial predictive maintenance, and anomaly detection, where opaque predictions

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can have substantial consequences. A growing body of work addresses the distinct interpretability requirements of these applications and the limitations of generic XAI methods in domain-specific decisions.

In healthcare and clinical settings, interpretability requirements extend beyond simple feature attribution to temporal reasoning at multiple granularities. An ICU user may need to know not only that a physiological variable mattered, but whether its recent deterioration, sustained elevation, or interaction with another signal drove the risk estimate. Badhon et al. (2025) introduced temporal concept tracking for ICU mortality prediction, addressing when clinically relevant patterns arise as well as which features are involved. Raab et al. (2022) developed an XAI workflow for epilepsy detection from EEG data that maps SHAP outputs across spectral, spatial, and temporal dimensions; its clinician study provides application-specific evidence about perceived usefulness and decision support. Earlier work reviewed by Vollert et al. (2021) highlights a related gap between generic post-hoc methods and predictive-maintenance workflows. In that setting, the timing and pattern of a highlighted sensor help connect the explanation to an inspection, maintenance, or shutdown decision. These examples motivate domain-specific interfaces and evaluation of whether an explanation helps users identify actionable failure modes.

Anomaly detection provides adjacent-task examples in which operators inspect why a segment was flagged and whether the event resembles an isolated spike, a persistent drift, or a change in periodic behavior. The Anomaly Transformer Xu et al. (2021) uses an "association discrepancy" between point-wise and series-wise attention as its detection criterion and exposes that discrepancy for inspection. CATCH Wu et al. (2025) provides frequency-band localization, helping users distinguish an anomaly concentrated in a particular periodic component from a broadband change. For forecasting, these mechanisms can be adapted by linking the highlighted temporal or spectral evidence to changes in a future prediction through task-specific perturbation tests.

Beyond model-specific interpretability, time-series explanations also require systematic evaluation. Leung et al. (2023) proposed WinIT, a temporally aware feature-importance method that accounts for lagged effects rather than treating every time step independently. Wang (2024) contributed sensitivity-based protocols using Feature Ablation, Occlusion, and Morris sensitivity with synthetic ground truth. Together, these methods offer concrete ways to evaluate time-series explanations beyond ad-hoc qualitative examples.

5.5 The Causality Discovery Landscape

The integration of causality into forecasting models is underpinned by a rich body of work on causal discovery from time series. Assaad et al. (2022) categorize these approaches into Granger-causal, constraint-based, noise-based, score-based, and hybrid methods, and emphasize persistent challenges such as unobserved confounding and the distinction between instantaneous and lagged effects. From a deep-learning perspective, Löwe et al. (2020) introduced amortized causal discovery, which separates causal structure from temporal dynamics and learns to infer structures across related

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datasets. Amortization can reduce repeated discovery cost, although the relationship between computational efficiency and causal fidelity remains an open question.

5.6 Foundation Models and LLM-based Forecasting

The success of large language models (LLMs) has inspired a rapidly growing line of work on foundation-style time-series forecasting. These methods introduce a different point in the efficiency–explainability landscape: they can support zero-shot or few-shot transfer and provide a natural-language interface, while also bringing substantial pre-training and inference costs. Their human-readable rationales can support communication, and grounding or perturbation tests can assess how closely the text reflects the computation that produced a forecast.

5.6.1 Architectural Approaches

Three main strategies have emerged Liang et al. (2024). *Prompt-based adaptation* represents time series as tokenized numerical sequences and relies on in-context pattern completion without changing model weights. LLMTime Gruver et al. (2023), for example, encodes scaled numerical values as text tokens and obtains a forecast from the continuation distribution of a pre-trained language model. This route requires no task-specific gradient update, but accuracy and cost depend on numerical formatting, context length, sampling, and the language model used. *Cross-modal alignment* retains a frozen language backbone while training lightweight components that project temporal representations into its token space. Time-LLM Jin et al. (2023) reprograms temporal patches through learned text prototypes, whereas GPT4TS Zhou et al. (2023) trains the input embedding and output projection around frozen Transformer layers. Freezing most weights reduces adaptation cost and can be useful when target data are limited, although the full backbone remains active at inference.

The third strategy develops *time-series-native foundation models* through pre-training on large and diverse temporal corpora. TimeGPT-1 Garza et al. (2023), Chronos Ansari et al. (2024), TimesFM, and Moirai use different tokenization, probabilistic output, multi-resolution, and multivariate strategies to support transfer across datasets. Unlike language-model repurposing, their representations and training objectives are designed around temporal values, covariates, and forecast horizons. They broaden the range of settings in which a forecaster can be reused and can reduce the need to train a separate model for every dataset. At present, however, most expose less fine-grained structural information than frequency-domain or basis-expansion methods; pre-training scale and transfer ability should therefore be considered separately from the form and fidelity of the explanations they provide.

The meaning of zero-shot or cross-domain efficiency depends on the information available to the model. Different studies use "zero-shot" to mean no target-series fitting, no dataset-specific tuning, or no gradient update, and a pre-training corpus may overlap the benchmark domain even when the exact test series is held out. Context length, numerical tokenization, prompt selection, probabilistic sampling, and

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API access can further change both accuracy and cost. Comparisons are most informative when they state the inference-time information and distinguish pre-training, adaptation, and per-forecast computation.

5.6.2 Efficiency-Explainability Analysis

From an efficiency perspective, foundation-style methods span several stages. Large Transformer backbones retain substantial parameter memory and may preserve quadratic attention in context length; lightweight adapters reduce trainable parameters while the full frozen backbone remains active at inference. Time-series-native foundation models use different parameter scales and tokenization strategies, so latency is best measured together with context length, batching, hardware, software, and decoding conditions.

LLM-based methods can also produce *natural-language rationales*. For example, a prompt may elicit a statement that a forecast extrapolates a weekly seasonal pattern and an upward trend. Such text can support communication, while perturbation, consistency, and model-internal grounding tests determine whether it traces the computation that produced the forecast. Under the proposed rubric, a rationale begins at Level 1 and can advance when these tests provide quantitative temporal, variate, or pattern-level evidence.

Critical Analysis: Conditional Deployment Trade-offs.

Foundation and LLM-based forecasters exchange task-specific training for pre-training, adaptation, and inference costs that vary considerably by model and deployment. They are particularly attractive when zero-shot or few-shot transfer is important or when a natural-language interface supports communication. In latency- or memory-constrained settings, their practical use may depend on distillation, quantization, or a smaller backbone. Grounding analyses can connect natural-language rationales to the temporal and variate processing that produced the forecast.

6 The Path Forward: Challenges, Directions, and Best Practices

The survey of contemporary and emerging MTS forecasting models reveals a rapidly developing field, but also one marked by fragmentation in evaluation, explanation reliability, and deployment evidence. Studies often use different forecasting protocols, the behavioral support for architecture-internal signals varies, and deployment reports rarely measure the joint cost of prediction and explanation. Building on these observations, this section outlines the most pressing open challenges, identifies promising research directions, and offers practical guidance for selecting and reporting forecasting models under different operational requirements.

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Table 10 Emerging Paradigms for Efficiency-Explainability Trade-off

Model/Paradigm	Year	Complexity	Params	Interpretability Strategy	Trade-off Resolution
Intrinsically Interpretable Architectures by Design					
N-BEATS Oreshkin et al. (2019)	2020	$O(L)$	∼5M	Basis expand (poly/Fourier)	Explicit basis decomposition
N-HiTS Challu et al. (2022)	2022	$O(L)$	∼1M	Hierarchical interp + multi-rate	Hierarchical components; study-specific speedup
CPformer Ma et al. (2025)	2025	$O(L^2)$	∼15M	5 domain-agnostic concepts + physics loss	Concept-aligned bottleneck; study-specific gains
RCL (Mamba) Yan et al. (2025)	2025	$O(L)$	—	Token-level contrastive pre-training	Focus Rate/Memory Entropy metrics; zero inference overhead
Causality-Aware Forecasting: From Correlation to Causation					
CUTS Cheng et al. (2023)	2023	$O(L^2)$	—	Granger + causal graph discovery	Graph recovery under stated assumptions
CUTS+ Cheng et al. (2023)	2023	$O(L^2)$	—	SCM integration	Counterfactual analysis under an SCM
Causal GNN Gong et al. (2023)	2023	$O(N^2L)$	—	Causal graph → GNN prop	Graph-constrained message passing
Adaptive Mechanisms for Non-Stationary Series					
Non-stat. Transf Liu et al. (2022)	2022	$O(L^2)$	∼30M	De-stat Attn (re-inject stats)	Moment adaptation; study-specific gain
Koopa Liu et al. (2023)	2023	$O(L \log L)$	—	Koopman operator (spectral dyn)	Interpretable modes; time-varying
FourierGNN Yi et al. (2023a)	2023	$O(L \log L)$	—	Fourier + GNN adaptive freq	Dynamic freq-band attn; evolving
Hybrid and Modular Designs: No One-Size-Fits-All					
ModernTCN Luo and Wang (2024)	2024	$O(L)$	∼2M	Res conv + decomp + patching	Modular: plug-and-play blocks
SOFTS Han et al. (2024)	2024	$O(L)$	∼0.5M	Stochastic Fourier + Online	Concept drift; online updates

Note: This table synthesizes models from Sections 4 and 5 that explicitly target the efficiency–explainability trade-off. Numerical gains retain the evaluation protocols reported by their sources. **Intrinsic architectures** (N-BEATS, N-HiTS) expose objects through structural constraints; **causality-aware methods** (CUTS, Causal GNN) add discovery costs and assumptions; **adaptive mechanisms** (Non-stationary Transformer, Koopa) respond to distributional shifts; and **hybrid designs** (ModernTCN, SOFTS) combine complementary modules. The final column summarizes how each paradigm shifts or conditionally relaxes the trade-off.

6.1 Open Challenges: Bridging the Gaps in Current Research

Reliable comparison of the next generation of architectures depends on progress in protocol comparability, explanation reliability, uncertainty, and the coverage of efficient channel-independent models.

6.1.1 Comparable Evaluation Protocols

MTS explanation studies frequently use custom metrics or qualitative cases, which limits direct comparison Rojat et al. (2021); Arsenault et al. (2024). A useful benchmark accounts for temporal dependence, high dimensionality, and correlated variates while evaluating several distinct properties: *fidelity* to model behavior, *robustness* to small or distribution-preserving perturbations, and *human utility* for domain users. Reporting these properties separately preserves their distinct meanings.

Efforts such as fev-bench Olivares et al. (2024) demonstrate confidence intervals, win-rate analyses, and aggregation across datasets and random seeds for forecasting accuracy. Extending this style of reporting to attention faithfulness, explanation stability, and human utility remains an open need. We recommend reporting variability and statistical uncertainty alongside mean performance, while recognizing that the appropriate validation method depends on the explanation object and application.

A comparable explanation protocol identifies the forecast and explanation targets, the unit of perturbation, the reference distribution, and the aggregation rule across samples, horizons, and seeds. Grouped or conditional masks can preserve temporal autocorrelation, correlation-aware perturbations can support variate explanations, and causal explanations can state their identification assumptions. These details reveal

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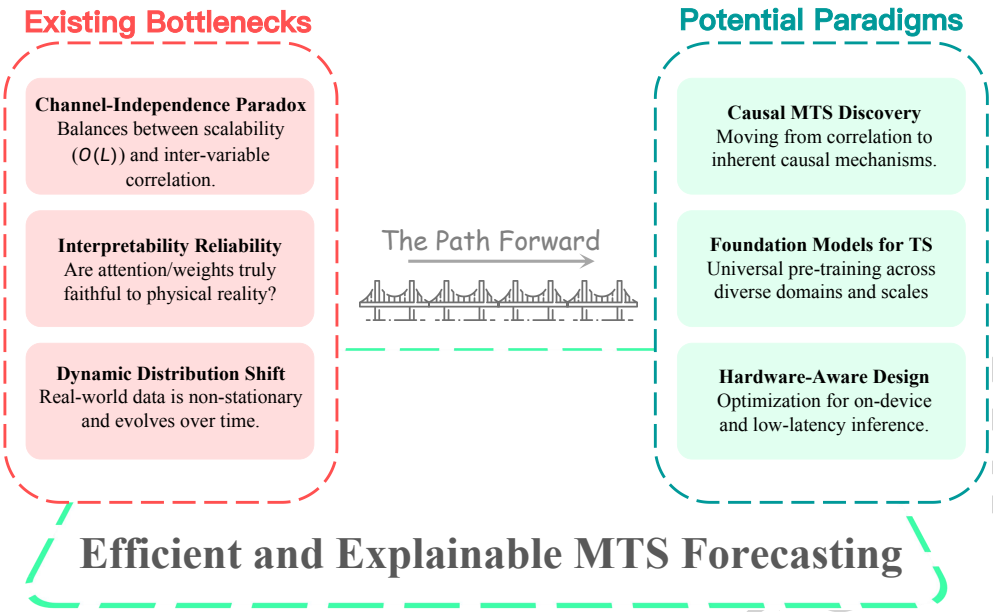


Fig. 8 Mapping representative bottlenecks and candidate directions for trustworthy MTS forecasting. The figure highlights the channel-independence paradox, explanation-reliability concerns, and distribution-shift challenges, alongside possible directions such as causal discovery, foundation-style time-series modeling, and hardware-aware deployment.

whether two reported scores measure the same property even when the studies share datasets and code.

Moreover, large-scale benchmarking studies such as BasicTS+ Shao et al. (2024) show how strongly dataset heterogeneity can affect model selection: no single architecture dominates across all evaluated settings. The trade-off between predictive accuracy, efficiency, and interpretability is therefore contingent on properties such as periodicity, non-stationarity, and inter-variable correlation. These characteristics should inform the choice of a model whose explanatory and operational properties fit the application. Beyond protocol comparability, the reliability of commonly used explanation methods remains a central concern.

6.1.2 Assessing the Faithfulness of Saliency-based Explanations

Saliency maps and attention weights are often used as proxies for feature importance because they are easy to visualize. Prior work reports cases in which attention differs from gradient or perturbation-based importance and changes while the prediction remains stable Bastings and Filippova (2020). Attention describes where information is aggregated; perturbation, stability, and sensitivity tests show how that aggregation relates to forecast changes. This distinction is especially important in MTS settings with strong correlation and temporal redundancy. This evidence motivates a more explicit analysis of the relationship between model performance and explanation quality.

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6.1.3 Quantifying the Trade-off: Moving Beyond a Qualitative Dichotomy

Efficiency and explainability are often framed as a choice between powerful black boxes and simple transparent models, but both properties are multidimensional. The framework in Section 3 separates computational measures from explanation scope, faithfulness, and overhead. Collapsing these dimensions into a single score would hide differences in deployment cost and in the questions a model can answer. A more informative direction is to report architecture- and application-specific Pareto relationships, allowing practitioners to decide which compromises are acceptable for a particular task.

6.1.4 From Point Forecasts to Probabilistic Understanding

While this survey has primarily focused on models that generate deterministic point forecasts, extending explanation methods to probabilistic forecasting is an important direction Zhang et al. (2025). In high-stakes domains such as clinical decision support or energy grid management, calibrated prediction intervals or full predictive distributions provide essential information about risk and confidence. Most existing XAI techniques explain deterministic outputs, leaving open how different inputs contribute to predictive variance or why a model is uncertain. Promising approaches include conformal prediction, interpretation of Bayesian neural network posteriors, and ensemble-based analysis of forecast variance. Probabilistic MTS explainability can also report calibration-oriented metrics such as CRPS, pinball loss, prediction-interval coverage probability, interval sharpness, and reliability curves alongside average accuracy.

MaGNet-BN provides a forecasting-oriented example by combining Bayesian embeddings and Markov transitions with long-horizon probabilistic prediction in dynamic graph environments Qu and Ma (2025). Related dynamic-community studies, including Bio-RegNet and TriMeta-BFNet, combine Bayesian uncertainty with stability or frequency diagnostics Ma et al. (2026); Qu et al. (2026c). Although these latter methods address adjacent tasks, their treatment of uncertainty offers useful methodological ideas for MTS forecasting that should be evaluated through task-matched calibration and faithfulness tests.

6.1.5 The Channel-Independence Paradox

A recurring theme is a channel-independence paradox. DLinear, FITS, OLinear, and channel-independent N-BEATS obtain low temporal cost and inspectable temporal or pattern components partly because they process each variate independently. ICU monitoring, financial risk analysis, and industrial diagnosis, however, often ask which variable drove the forecast. Answering this question requires a channel-dependent mechanism or a post-hoc method such as SHAP, permutation tests, or occlusion. These methods add repeated model evaluations and require care when variates are strongly correlated. Lightweight channel-dependent designs and low-overhead variate attribution can therefore complement the efficient temporal and pattern structure of

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CI models. Together, these challenges define the methodological foundation on which future advances in explainable time-series forecasting will depend.

6.2 Future Research Directions: Charting the Next Wave of Innovation

Progress in evaluation, reliability, and uncertainty will support forecasting models that are not only accurate and efficient but also more useful in decision-making. The following directions appear particularly promising.

6.2.1 From Correlation to Causation: The Pursuit of Actionable Explanations

Most current explainability methods are correlational: they identify input features associated with an output. Decisions involving interventions additionally call for causal questions and explicit identification assumptions. As discussed in Section 5, causality-aware forecasting provides a basis for this extension Gong et al. (2023). Three technical areas are particularly relevant:

1. **Causal Discovery Integration:** Building models that not only forecast but also simultaneously infer a partial causal graph from time series data, using the learned graph to structure both the predictions and the explanations.
2. **Counterfactual XAI:** Extending "which feature was important?" with questions such as "what is the minimal feasible change to an input series that would alter the forecast?" Wachter et al. (2017).
3. **Interventional Robustness Evaluation:** Assessing explanation stability under formal causal interventions within the data-generating process, alongside standard data perturbations.

Causal reasoning can provide more actionable insights while adding graph-discovery, intervention, and counterfactual-computation costs. This motivates architectures that expose useful explanatory structure during forecasting.

6.2.2 Efficient by Design: Native Explainability for Resource-Constrained Environments

Post-hoc explainability is flexible but incurs additional computational cost and requires method-specific faithfulness evaluation. A complementary long-term direction is to design models with *native explainability*, where forecasting components are themselves interpretable. Block-structured models such as N-BEATS Oreshkin et al. (2019) and effective linear architectures Zeng et al. (2023) provide useful foundations. The diversity of MTS tasks favors flexible systems that adapt their forecasting and explanation mechanisms to the data and deployment requirements. Technical directions include:

1. **Architecture-XAI Co-design:** Using interpretability constraints, such as monotonicity and disentanglement, as inputs to architecture design.

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2. **Quantization-Aware Explainability:** Researching how model compression and quantization techniques, which are essential for deployment on resource-constrained edge devices, impact model explanations. The goal is to develop methods whose fidelity is robust to reduced numerical precision.
3. **Self-Explaining Linear Models:** Extending current linear models to highlight influential lookback windows, cross-channel dependencies, and detected trend or seasonal components during the forward pass.

6.2.3 Latency-Bounded Explainability for Real-Time Deployment

Explanation cost is distinct from forecasting cost. Adding SHAP Lundberg and Lee (2017b), LIME, Integrated Gradients, or a counterfactual search to a lightweight forecaster can require additional gradient evaluations, perturbation samples, surrogate fitting, or optimization iterations. The resulting latency depends on the explainer configuration, number of variates and lags, batching, software, and hardware. End-to-end measurements of forecasting and explanation on the target platform therefore provide the relevant deployment evidence.

The explanation budget should also be tied to the decision deadline and required granularity. A coarse channel ranking generated for periodic review has a different latency requirement from a per-timestep explanation delivered with every alarm. Approximate or cached explanations can be useful when their fidelity and stability are measured at the selected sampling or search budget. Joint reporting of forecast quality, explanation faithfulness, and end-to-end cost then reveals the operational trade-off.

This motivates bounded-latency explanation mechanisms, including structured cross-channel components that expose inspectable signals during the forward pass, sparse gating, and hardware-aware attribution whose granularity can be adjusted to the deployment budget. Behavioral tests can be integrated into their evaluation to connect these internal signals to the forecast.

6.2.4 Addressing the Frequency-Domain Frontier

The recent comprehensive survey by Zhang et al. (2025) identifies three critical open challenges that remain largely unaddressed by the explainability community:

1. **Causal Structure Preservation in Spectral Transforms:** Standard Fourier and wavelet transforms use information across the analysis window, so causal explanations require transforms and masks that preserve temporal ordering while retaining frequency-domain efficiency.
2. **Uncertainty Quantification in Learned Frequency Representations:** Bayesian or ensemble methods can attach confidence information to spectral components and help practitioners assess the reliability of frequency-based explanations.
3. **Topology-Aware Analysis for Non-Euclidean Data:** Many real-world MTS exhibit graph-structured dependencies, including sensor networks and traffic systems. Spectral graph methods can extend frequency-domain interpretation to this non-Euclidean structure.

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Related temporal-graph and sequence-generation studies illustrate several robustness mechanisms: Fourier–Markov diffusion for class-preserving time-series generation, temporal-graph community tracking, fractional/Fourier filtering with Lyapunov analysis, perturbation-bounded spectral regularization, and cross-snapshot stability regularization Ma et al. (2026a,b); Qu and Ma (2026); Qu et al. (2026a,b). Although developed for generation and clustering, these mechanisms motivate forecasting analyses of spectral sensitivity, perturbation-induced drift, and temporal consistency alongside forecast error.

6.2.5 Unlocking the Black Box of State Space Models

SSMs combine linear temporal scaling with strong forecasting results in several recent studies, yet their hidden states, selective gates, kernels, scan dynamics, memory decay, and input-dependent transitions remain difficult to connect to individual forecasts. This creates an explanation gap even when their predictive and computational properties are favorable.

Initial efforts in this direction are emerging. Metrics introduced with Repetitive Contrastive Learning (RCL), including *Focus Rate* and *Memory Entropy*, offer a starting point for quantifying how SSMs allocate and retain information over time Yan et al. (2025). Future state-space interpretability should also account for the interaction between recurrent scans and cross-variate tokenization. Relevant internal objects include hidden states, selective gates, convolution kernels, scan dynamics, memory decay rates, and input-dependent state transitions. Existing metrics provide useful summaries but not yet the temporal and variate attribution available for some other architectures.

Future work could address this gap by (1) translating learned state-space matrices (A, B, C, D) into understandable feature transformations and temporal dynamics; (2) adapting perturbation or surrogate methods to the recurrent structure of SSMs; and (3) combining efficient state-space processing with an interpretable attention or decomposition layer. These directions could preserve long-range efficiency while making state dynamics easier to connect to individual forecasts.

6.2.6 Hybrid and Adaptive Paradigms: No One-Size-Fits-All

The diversity of time-series characteristics—from stationary to highly non-stationary Liu et al. (2022), and from sparse to dense—motivates hybrid and adaptive paradigms. These systems can adjust their processing and explanation modalities to the data at hand. Key technical pathways include:

1. **Dynamic Routing Networks:** Designing models with multiple specialized processing blocks (e.g., a spectral block for seasonal data, a Transformer for long-range dependencies) and a learned routing mechanism that dynamically allocates computational resources based on the input series' statistical properties Shazeer et al. (2017).
2. **Explanation on Demand:** Creating multi-modal systems that can operate in a highly efficient, black-box mode for routine forecasts but can generate

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detailed, computationally intensive explanations when explicitly queried by a human operator for a specific, critical prediction.

3. **Symbolic Knowledge Fusion:** Integrating sub-symbolic deep learning models with symbolic knowledge bases (e.g., expert-defined rules, physical constraints). This allows the model to ground its explanations in established domain knowledge, making them more verifiable and trustworthy.

6.3 Practical Recommendations: A Guide for High-Stakes Deployment

Alongside these research directions, practitioners need clear guidance for navigating the current model landscape in high-stakes environments. This section translates the preceding analysis into recommendations for practitioners navigating the efficiency–explainability trade-off.

6.3.1 A Decision Framework for Model Selection in Practice

The model families in Sections 4 and 5 can be screened through a structured sequence of operational and explanatory requirements:

1. **Start with Governance and Oversight Requirements:** Identify the documentation, oversight, and explanation scope required by the application and applicable governance process Wachter et al. (2017). A need for temporal or pattern inspection may favor structured linear or decomposition candidates, whereas cross-variate attribution calls for a channel-dependent model or a separately validated post-hoc method. Computational cost and faithfulness can then be assessed under the deployment protocol.
2. **Assess Variate Attribution Requirements:** Before evaluating computational budgets, determine whether the application requires cross-variate explanations. In predictive maintenance or clinical decision support, questions such as "which sensor or indicator is driving this anomaly?" call for a channel-dependent mechanism and can take priority over temporal efficiency alone.
3. **Assess the Computational Budget:** What are the strict constraints on inference latency, training time, and available hardware? For edge deployments or real-time control systems, lightweight linear models (e.g., DLinear, FITS) or highly optimized architectures (e.g., ModernTCN Luo and Wang (2024), SOFTS Han et al. (2024)) are strong candidates due to their minimal parameter footprint and linear complexity. For offline analysis with large budgets, more complex models like FEDformer may be justified.
4. **Analyze Problem Complexity and Data Characteristics:** Is the primary challenge capturing long-range dependencies, dealing with severe non-stationarity, or modeling complex inter-variable relationships? A careful analysis of the data's properties should guide the choice. Transformers excel at long dependencies, while models explicitly designed to handle distribution shifts (e.g., Non-stationary Transformer Liu et al. (2022), Koopa Liu et al. (2023)) are better suited for non-stationary data. For industrial applications requiring a balance of

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cross-domain scalability and interpretability, the Temporal Fusion Transformer (TFT) Lim et al. (2021) remains a robust choice due to its gated residual networks and variable selection components, although its $O(L^2)$ attention mechanism necessitates careful hardware provisioning for long sequences.

5. **Consider Maintenance and Expertise Overhead:** Does the operational team possess the deep learning expertise required to maintain, debug, and retrain a complex model? A simpler, more robust model that is easier to understand and manage may provide greater long-term value than a marginally more accurate but brittle and opaque alternative.

Table 11 summarizes this decision process across representative deployment scenarios. The suitable family changes with the dominant requirement, from low latency to cross-variate attribution, pattern decomposition, or calibrated uncertainty.

The guide begins with the forecast target and explanation scope. Cross-variate attribution, for example, narrows the candidates to models or explainers that exchange information across channels. The remaining models can then be compared under compatible data splits, horizons, preprocessing, and tuning budgets, followed by measurement of prediction and explanation cost on the intended hardware. Finally, the selected explanatory object can be evaluated against model behavior and, where relevant, domain-user needs. This sequence aligns model choice with the operational requirement.

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Table 11 Scenario-oriented model selection guide for MTS forecasting, organized by explanatory requirements and deployment constraints.

Scenario	Primary constraint	Suitable model families	Explanation mode	Main caveat
Low-latency edge deployment	Small parameter budget, bounded memory, fast inference	Linear and lightweight temporal models such as DLinear, FITS, ModernTCN, and SOFTS Zeng et al. (2023); Xu et al. (2024); Luo and Wang (2024); Han et al. (2024)	Temporal saliency and pattern decomposition	Channel-independent backbones require an additional mechanism for cross-variate attribution.
High-dimensional sensor networks	Large N, cross-channel interactions, possible graph structure	Variate-aware or graph-aware architectures such as iTransformer, Crossformer, and topology-aware spectral models Liu et al. (2024); Zhang and Yan (2023); Zhang et al. (2025)	Variate attribution or relation-level explanation	$O(N^2)$ attention can become expensive; perturbation and stability tests strengthen attention or graph explanations.
Need for cross-variate attribution	Identify which sensor, asset, or physiological channel drives the forecast	Channel-dependent attention, gating, or post-hoc perturbation over variates Zhang and Yan (2023); Liu et al. (2024); Lundberg and Lee (2017b)	Validated variate ranking and cross-channel saliency	Post-hoc attribution increases latency and requires correlation-aware perturbations.
Need for pattern decomposition	Explain trend, seasonality, residual, or frequency components	Decomposition and frequency-domain models such as N-BEATS, N-HiTS, FITS, FEDformer, and Ister Oreshkin et al. (2019); Challu et al. (2022); Xu et al. (2024); Zhou et al. (2022); Cao et al. (2024)	Pattern-level explanation through bases, filters, or components	The decomposition scope depends on the basis and channel-mixing design.
Need for uncertainty-aware decisions	Calibrated risk, prediction intervals, or downstream triage	Probabilistic forecasting, conformal prediction, Bayesian or ensemble extensions	Interval calibration, uncertainty attribution, and risk-sensitive reporting	This extends the three-pillar framework from point forecasts to predictive uncertainty.

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6.3.2 Beyond the Algorithm: Integrating Explanations into Human-in-the-Loop Workflows

Model selection is followed by the integration of explanatory outputs into the decision workflow. In high-stakes applications, XAI supports experts by presenting model behavior in a form that can be examined alongside domain evidence. This places the human–computer interface and decision workflow alongside the explanation algorithm itself. Useful practices include:

1. **Use Explanations for Hypothesis Generation:** Saliency maps and related explanations can help domain experts identify previously unknown leading indicators, detect anomalous sensor behavior, and refine their mental models of the system.
2. **Build Calibrated Trust:** Explanations should reveal a model's failures as well as its successes. Workflows can help operators understand when to rely on a forecast and when to override it, particularly when an explanation indicates dependence on a known spurious feature.
3. **Design Interactive Interfaces:** Static explanation dashboards are of limited use. The most effective systems will provide interactive interfaces that allow users to drill down into forecasts, ask "what-if" counterfactual questions, and compare the reasoning of multiple models side-by-side.

6.3.3 A Proposed Reporting Checklist for MTS Forecasting Research

To support transparent and reproducible comparison, we propose the following compact reporting checklist for future studies:

1. **Dataset and Protocol:** Report datasets, splits, lookback windows, forecasting horizons, preprocessing, and evaluation metrics.
2. **Seeds and Statistics:** Report the number of random seeds, aggregation procedure, variability, and confidence intervals where applicable.
3. **Hardware and Software:** Specify the processor or accelerator, memory, framework, software versions, and relevant runtime settings.
4. **Efficiency:** Report L- and N-dimension scaling, parameter count, latency, and peak memory under a stated measurement protocol.
5. **Explanation Scope and Mechanism:** State whether the method addresses temporal, variate, or pattern explanations and whether the explanatory object is intrinsic or post-hoc.
6. **Faithfulness and Reproducibility:** Report the validation metric and perturbation protocol for explanation claims, together with the availability of code, configurations, and data-processing details.

6.4 Scope and Limitations

This survey provides a taxonomy-driven synthesis of MTS forecasting studies conducted with heterogeneous datasets, horizons, preprocessing choices, hardware, and hyperparameter budgets. The evidence tables use these studies to identify recurring

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architectural patterns and evaluation gaps, while controlled model rankings call for matched experimental protocols.

The Level 0–3 protocol and the ETTh1 case study in Table 5 operationalize attention-faithfulness evaluation for one dataset family, one horizon, and two models. Broader calibration can combine synthetic ground truth, multi-dataset evaluation, additional checkpoints and architectures, and human-utility studies.

Our primary framework targets deterministic point forecasting. Probabilistic forecasting, calibration, conformal prediction, and uncertainty attribution form an important extension for high-stakes settings in which calibrated risk may be as important as point accuracy.

The corpus design favors peer-reviewed venues and influential arXiv preprints, which gives recent 2024–2025 work less time to demonstrate influence through citations. Snowballing and expert screening improve coverage, and periodic updates will be valuable as foundation models, state-space models, and graph-based MTS forecasting mature.

Finally, in regulated settings, model explanations contribute to auditability and human oversight as part of a broader process that includes documentation, risk management, institutional controls, and domain-specific governance.

7 Conclusion

This survey has examined the evolving landscape of deep learning for multivariate time series forecasting through the dual requirements of computational efficiency and model explainability. Its primary contribution is a systematic synthesis of this literature, developed through three connected contributions.

First, we organize the literature through a dual-axis view of sequence-length and variate-dimension scaling and a three-question taxonomy of Temporal Importance, Variate Importance, and Pattern Decomposition. The accompanying Level 0–3 rubric and Explanation Overhead axis provide supporting tools for comparing how different model families expose and validate explanatory information.

Second, our review of Transformer, frequency-domain, linear, state-space, graph-based, and foundation-style approaches shows that the efficiency–explainability relationship is architecture-dependent rather than uniformly antagonistic. Channel-independent linear models and block-structured architectures can combine low computational cost with temporal or pattern-level transparency. At the same time, the absence of native cross-variate attribution limits the scope of that transparency, a distinction represented by the Level $3^{\dagger}$ designation. Attention maps, learned graphs, decomposition components, and natural-language rationales likewise expose different objects and require different forms of validation.

Third, the synthesis identifies three continuing challenges: developing comparable benchmarks for explanation quality, connecting state-space dynamics to individual forecasts, and advancing from correlational attribution toward causal explanations supported by explicit assumptions. Progress on these questions requires reporting predictive accuracy, computational cost, explanation mechanism, and faithfulness as distinct but connected dimensions.

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The path forward lies in the joint development of efficient forecasting architectures and explanations that are appropriate to their intended decisions. We hope that the taxonomy, evidence catalog, model-selection guide, and reporting considerations assembled in this survey provide researchers with a useful basis for comparison and help practitioners choose forecasting systems whose computational and explanatory properties match their deployment requirements.

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